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SENTIENCE AND SUFFERING

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2.1 Sentience, wellbeing, suffering, and moral standing

In this chapter, we begin to investigate what suffering is. This is necessary to ascertain what it takes for AI systems to be capable of suffering, and how likely AI suffering is. The main thesis of this chapter is that sentience is a sufficient, even though perhaps not a necessary, ground for the capacity for suffering.[1] All sentient beings can suffer. Before developing, motivating, and defending this thesis, I first lay some conceptual foundations. So, this section first provides definitions of the key terms relevant to suffering and moral relevance. While these definitions are in accordance with the philosophical literature, they also have a stipulative element since many terms are commonly used in ambiguous and conflicting ways.

The first relevant concept is *consciousness*. In this book, when I use the term consciousness without further clarification, I mean *phenomenal* consciousness. Mental states are phenomenally conscious if and only if they involve a subjective experiential feel. In humans, states like smelling coffee or seeing a red rose are often phenomenally conscious, while dreamless sleep or synthesizing glucose occur non-phenomenally. This characterization pertains to how the state is experienced from the first-person perspective, respectively, "what it's like" (Nagel 1974) to be in such a state. A being, e.g., an AI, possesses phenomenal consciousness if and only if it undergoes states which are phenomenally conscious.

Sentience is the capacity to have (phenomenally) conscious experiences which have a valence (Crump et al. 2022). An experience is valenced if and only if it feels good or bad. Emotions, bodily sensations, and moods are 12typically valenced. For instance, joy and relief have a positive valence and fear, pain, and boredom have a negative valence. By contrast, a typical visual experience may have no valence. It does not feel good or bad to see an ordinary apple, although this is still a conscious experience with a specific character. What it's like to see an apple is different from what it's like to see a banana. "Consciousness" and "sentience" are both concepts which refer to the capacity for certain kinds of mental states. The next concepts refer to certain ethical properties.

Wellbeing is what is intrinsically good (or bad) for the very being whose wellbeing it is (Crisp 2021; Dung 2023). So, wellbeing is valuable. The relevant value here is intrinsic (or non-instrumental). This means that the value distinctive of wellbeing is independent of its relation to other states it may bring

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about. Wellbeing is valuable, i.e., good for someone, even if having high wellbeing does not cause other valuable things. Moreover, the value of wellbeing is conceptually independent from whether a life is morally good. The value distinctive of wellbeing only concerns whether things are good for the being in question, not whether a person is virtuous or morally upstanding. So, it is conceivable that a mass murderer has high wellbeing.

Importantly, negative wellbeing is what is intrinsically bad for the person who is subject to it. When philosophers discuss what wellbeing is, they are asking about the property (or the set of properties) which determines to what extent someone's life is intrinsically and non-morally good or bad; i.e., they are asking which property wellbeing consists in or is constituted by. I will call this property a "ground" or "constituent" of wellbeing. Happiness may be one example, since happiness seems to be good for someone, even if it does not have any further positive effects (so intrinsically). This contrasts with, for example, money. Having money is only good for one because it can be used to get other things, not intrinsically. I will use the term "welfare" as a synonym for "wellbeing".

I will equate suffering here with negative wellbeing. That is, *suffering* means being in states which are intrinsically bad for you. This deviates from how the term suffering is often used because it means that suffering does not have to consist in experiences. My definition implies that it is conceptually possible that someone suffers without having unpleasant conscious experiences. For example, imagine an experience machine (Nozick 1974) which supplies you with all the pleasurable experiences in the world by mechanically stimulating your brain, but it is all an illusion (without you knowing or experiencing that it is illusory). Many philosophers would say that entering the machine is bad for you because it deprives you of virtually all mind-independent goods: knowledge of the external world, real interactions with your loved ones, etc. Our definition leaves it open 13that being in the machine might be bad for you and thus constitute suffering. This is an untypical way of using the word "suffering" since it does not involve negative conscious experience.

This disadvantage is offset, however, by the utility of understanding suffering that way. For when we say that we want to prevent suffering risks, we should not limit the scope of our project at the outset. Ultimately, we should aim to prevent everything which is intrinsically bad for beings, be they conscious experiences or something else. So, we should understand suffering in this broader way. That being said, this terminological point is compatible with holding that, as a matter of fact, negative conscious experiences are all that is intrinsically bad for beings.[2] We will discuss this view later.

There are some much richer, and potentially ambivalent, conceptions of suffering in the literature. For example, Brady (2018) argues that human suffering is necessary for virtue and, as a consequence, for leading a flourishing life. For example, he claims, suffering causally promotes virtues such as bravery and humility and is constitutively involved in the expression of virtues such as compassion. I do not take a stand on these claims. However, on the definition of suffering presupposed here, it is inherently contradictory to say that positive wellbeing constitutively involves suffering. It is, of course, possible that previous states of negative wellbeing causally promote high wellbeing at a later point in time (e.g., previous unhappiness may make later happiness more likely). However, if negative emotions constitute flourishing and high wellbeing, then – on my notion – they do not constitute suffering. In my view, suffering is by definition non-instrumentally bad for the being which suffers.

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A being possesses *moral standing* if and only if it matters morally for its own sake, i.e., intrinsically (non-instrumentally) (Jaworska and Tannenbaum 2021). That is, if a being has moral standing, we have obligations to that being in virtue of its intrinsic properties, not because it matters to someone else. We ought to consider how our decisions affect beings with moral standing in our moral deliberations. The being can provide us with (non-instrumental) moral reasons to treat it in certain ways. For instance, we ought to consider whether our decisions harm or benefit other humans and weigh this in our decision-making. By contrast, whether our actions might destroy an object, e.g., a TV, is only relevant insofar as this object matters to beings with moral standing. The TV does not have moral standing; it is at best instrumentally relevant. Based on the notion of moral standing, we can also define *moral weight*: how much a being matters morally for its own sake.

The next section will begin the discussion of how these properties hang together.

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2.2 The sentience suffering connection

Let me introduce a last and simple definition: A *welfare subject* is a being which can have wellbeing. For instance, a rock is not a welfare subject, because nothing can be intrinsically good or bad for it, but a human is. I believe that being a welfare subject is equivalent to having moral standing. In other words, I believe that a being matters morally for its own sake if and only if things can be non-instrumentally good or bad for it. For simplicity, I will assume this view in this book. Yet the argument of this book does not strictly rely on this claim. All the arguments which I put forward in this book for the moral significance of certain properties could be framed in terms of either moral standing or welfare, without losing their plausibility. Thus, the independent claim that being a welfare subject and having moral standing are equivalent is not necessary for the argument. It does shorten the argument, however.[3]

The view that being a welfare subject is equivalent to having moral standing is highly plausible. If nothing can be (intrinsically) good or bad for a being, we cannot have intrinsic moral reasons to perform actions which are good or bad for it. What other kinds of intrinsic moral reasons could we have to treat this being in certain ways? It is not clear what the answer would be. Yet, if there are no possible non-instrumental pro tanto[4] moral reasons to behave toward a being in certain ways, it does not have moral standing. It does not figure (for its own sake) in our moral deliberation. If this argument is correct, then being a welfare subject is necessary for having moral standing.

It seems clear that, if certain things are intrinsically good or bad for beings, we can have – at least in some contexts – moral reasons to perform actions which are good for them. This is entailed by the view that we have reasons to increase wellbeing, which most ethical theories can agree on (Beauchamp 2019). On top of that, even views, e.g., Kantian ethics, which are skeptical of this principle, commonly accept the view that all welfare subjects have moral standing (Korsgaard 1996, 2018). If this argument is sound, then being a welfare subject is sufficient for having moral standing. In conjunction, we have strong reason to believe that being a welfare subject and having moral standing are equivalent.

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For simplicity, I assume that being a welfare subject is equivalent to being capable of suffering. Given our expansive concept of suffering, this means: All and only beings which can have wellbeing can have negative wellbeing. Note, however, that there is a theoretical possibility that beings could exist which are only capable of positive wellbeing, not of negative wellbeing, which would refute this claim. For example, we might create AI systems which are welfare subjects, but incapable of suffering, if we intentionally 15design AI systems this way as a response to AI suffering risk. Thus, it is useful to keep this simplifying assumption in mind.

Since equivalence is a transitive relation, this means that being a welfare subject, being capable of suffering, and having moral standing can all be treated as equivalent, at least in the present context. For this reason, I will often use these terms interchangeably in what follows. When I say that certain conditions are necessary or sufficient for having moral standing, they are also necessary or sufficient for being a welfare subject or being capable of suffering.

In the rest of this chapter, I will explore the connection between sentience and having moral standing. According to a common view, sentience is grounded in and necessary and sufficient for moral standing (Jaworska and Tannenbaum 2021; Königs 2025; Kriegel 2019; Singer 2011). Call this view *strong sentientism*: All and only sentient beings have moral standing, and they have moral standing in virtue of being sentient. If this view is correct, then to know whether we have moral reasons to help AI systems, we need to know whether they may be sentient. In this chapter, we will only examine one half of the sentientist thesis, namely, the claim that sentience is sufficient for moral standing. In Chapter 5, we will look at the contention that sentience is necessary for moral standing. Define *weak sentientism*: Sentience is a sufficient ground of moral standing. This view is neutral on whether sentience is necessary for moral standing. In other words, according to weak sentientism, if beings are sentient, then they have moral standing in virtue of being sentient, but beings may have moral standing without being sentient.

These formulations of strong and weak sentientism focus on moral standing, rather than being a welfare subject or being capable of suffering. This should alleviate worries of people who doubt the equivalence between moral standing, wellbeing, and suffering I have just argued for. Out of the three, the connection between sentience and moral standing is the least obvious. If one accepts that sentience is sufficient for moral standing, one should also accept that it is sufficient for being a welfare subject and being capable of suffering.

Strong sentientism is a popular, but controversial view. In the context of AI specifically, a variety of authors have argued that beings might have moral standing without sentience (Coeckelbergh 2010; Danaher 2020; Gunkel 2018, 2019; Roelofs 2022; Shepherd 2024; Shevlin 2021). By contrast, weak sentientism has very few detractors. In this section, I will put forward two arguments for weak sentientism. In the next section, I will defend weak sentientism against two objections.

Following Kagan (2019), Ladak (2023) proposes a “meta-criterion” for ascertaining which beings have moral standing. According to this criterion, 16a being has morally relevant interests, i.e., moral standing, when it has interests that “matter to it” or that it “cares about”.[5] Ladak provides the following argument for this principle:

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> When thinking about who should qualify for moral standing, we are asking which interests should matter in our moral decision making. But if an entity's interests do not matter to them, why should they matter in our moral decision making (for their own sake)? There seems to be no reason to take such interests into account. On the other hand, if an entity has interests that do matter to them, it seems clear to me that, if we want to act morally, we should take those interests into account.
>
> (*Ladak 2023*)

So, he appeals to two underlying principles: First, it is implausible that interests should matter to our moral decision-making (for their own sake) if they do not matter to the being whose interests they are. The second argument is less explicit, but it could be treated as the reverse of the first. This would give us the following principle: If the interests of a being matter to itself, then they should also matter to us (for their own sake).

Why should we accept these principles? The first principle is intuitive. If the being does not regard (in some sense) its own interests as important, it is strange that others should do so. This is because, when we honor our moral obligations to someone, we do it *for the sake of* this other being. It is questionable how we can reasonably act for the sake of another being if what we do does not matter to this being. In this case, we seem to make a mistake, imposing on the being attitudes it does not actually have.

One may object to this principle in the following way: Suppose a person is emotionally very distressed. It seems that we have a reason to remove this emotional distress if we can easily do so. Suppose further that the person is in a state which is often characterized as "self-destructive"; i.e., they seek out harm to themselves. One may describe this as a situation where a person does not care about their state of emotional distress, but we still have a moral reason to remove this distress. Hence, this would be a case of a morally relevant interest which does not matter to the person whose interest it is.

To reply to this counterexample, something's "mattering" to someone needs to be understood in a sufficiently broad sense. We must be able to say that the emotional distress still matters to the person, even though they do not attempt to avoid it. We will later clarify the principle so as to avoid counterexamples of this sort.

Let us for now move to the second principle. It was: If the interests of a being matter to itself, then they should also matter to us (for their own sake). 17One may support this principle with the following reasoning: If the interests of a being matter to it, then it would violate an ideal of impartiality to not have them matter to us. Singer (2011) argues for a principle of equal consideration of interests: Everyone's interests matter equally (relative to their strength), no matter whose interests they are. By analogy, one may say that a being's own interest mattering to it should have the same significance, irrespective of what being we are talking about. Arguably, with humans, we take the fact that an interest matters to a human as a sufficient reason to have it matter for us. If so, we should do the same for non-human beings.

This argument is not conclusive. One may object that, even in humans, having an interest which matters to oneself is not sufficient for having morally relevant interests, since some other condition (e.g., consciousness) is required. While this view is coherent, I favor the view that, in humans, having interests that matter to oneself is sufficient for moral standing and that sentience is something which contributes to interests mattering to oneself. Hence, I endorse Ladak's proposed meta-criterion here.

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Like Ladak, I hold that all sentient beings have interests that matter to themselves, thus qualifying for moral standing. At a minimum, they have an interest in experiencing pleasant states (e.g., joy) and not experiencing unpleasant states (e.g., pain). In addition, since sentience entails consciousness, there is a conscious subject to whom these states matter.

The idea is that valenced conscious states matter to beings in virtue of how they feel, not just in virtue of some cognitive attitude. Suppose a being does not have explicit pro-attitudes toward its conscious experiences; i.e., it does not desire to (not) have certain experiences. For instance, because it is incapable of metacognition (Gennaro 2009, 2012), i.e., the capacity to form thoughts or other representations about its own mental states. This may be the case because metacognition is often thought to be quite demanding, perhaps even limited to mammals (Carruthers 1998, 2007; though see: Smith and Washburn 2005). Nevertheless, the experience can still matter in virtue of how it feels. This provides us with an answer to the challenge of "self-destructive" people: Even if a person does not care about or tries to resolve their emotional distress in a more cognitive sense, reflected in their explicit preferences, the experience's feeling unpleasant is sufficient to say that it matters to the person.

So, it appears that the meta-criterion considered supports weak sentientism. For it seems plausible that if a being is sentient, then it has interests that matter to it. A second source of support for weak sentientism stems from considering particular cases (Dung 2022a). Employing the method of cases is common in debates on moral standing, wellbeing, and intrinsic value (Crisp 2021; Roelofs 2022). To apply this method, one has to create minimal contrasts: two situations which differ descriptively only 18 in the presence of a particular property. If those situations also differ in their ethical properties, then this shows that the particular property in question is ethically significant (Kriegel 2019; Levy 2014).

For instance: Imagine Kate, who is an ordinary, healthy person who goes to bed every night and sleeps until morning without interruption. Contrast this with another person, call her Anne, who is the same in all respects, except that she wakes up every night and is haunted for 20 minutes by excruciating pain, before falling back to sleep again (Kammerer 2019). It seems clear that the night is bad for Anne in a way in which it is not bad for Kate. For this reason, we have a reason to cure Anne from her pain problem if we can do so. This means: If we can help Anne without any cost to us and there are no counteracting moral reasons, then we should cure her. At the same time, there is no moral reason to do anything about the sleep of Kate. Hence, we have found an ethical difference between these cases.

What could explain this difference? The best explanation is that the second case involves strong pain experience, while the first one does not. Let us go through some alternative explanations. The mere fact that Anne wakes up every night cannot explain the ethical difference, as it – without the pain experience – is not (sufficiently) morally significant. To confirm this, suppose Kate also wakes up every night for 20 minutes, has experiences which are hedonically neutral (neither pleasant nor unpleasant), and then falls back to sleep. It seems clear that we have at least stronger moral reasons to cure Anne's sleeping problems than Kate's, since they involve pain experience.

Someone might say that pain experience is what explains the moral difference between these cases, but only indirectly. Anne's chronic pain might impair her functioning in all kinds of ways, maybe even traumatizing her. Perhaps these impairments, preventing her, for instance, from achieving some of her life goals, are what actually makes the moral difference between the cases. However, we can just modify the case again: Suppose Anne and Kate have no recollection of what they experience when they wake up

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at night. Moreover, all psychological and bodily traces of this wake up are erased every night so that their wake up has no cognitive or other effects. So, Anne does not know that she suffers from pain every night and her experiences cannot cause any relevant impairments. Nevertheless, I take it, we have a reason to cure her from her pain experiences. For she has to consciously experience these pains while they are present, even though the memory is erased later. While one may argue that erasing memories of the experiences makes them less bad, it still does not make them alright, i.e., morally neutral.

19Hence, even if we hold all factors between two situations except conscious experience constant, there can be differences in terms of our moral reasons between the situations. This suggests that the difference in conscious experience explains the difference between the situations. Hence, a being's conscious experiences can generate moral reasons to help it. This suggests that certain kinds of conscious experiences suffice for moral standing. There are two kinds of doubt one might have.

First, one may think that sentience is not sufficient for moral standing because not every kind of valenced conscious experience suffices to generate moral reasons. However, we can just vary the previous argument. It seems to me that the argument is similarly persuasive when changing the conscious experience of Kate from pain to another valenced state, such as fear or regret. In particular, positively valenced experiences also evoke the intuition that we have moral reasons to intervene. For instance, when Kate wakes up every night and has blissful experiences which she subsequently forgets, then we have a moral reason to enable her to continue having such experiences. The moral reason may be weaker than in the case of pain, but it still exists.

Second, one may think that valenced conscious experiences only provide us with moral reasons when they are the experiences of beings with moral standing. They do not make beings have moral standing. This move seems ad hoc to me. In the absence of an argument, I would assume that the set of facts which determines moral standing and the facts that determine our moral reasons toward beings with moral standing are the same.[6] According to this view, if someone has non-instrumental moral reasons to behave in a certain way toward a being, then this being has moral standing. Moral standing is not an antecedent and additional condition that must be satisfied for the possibility of such moral reasons; moral standing is entailed by this possibility.

To bolster the case that sentience suffices for moral standing, we can consult cases which are directly about moral standing. Consider a robot vacuum cleaner. Almost everyone agrees that it does not have moral standing. Now imagine, contrary to the facts, that this robot is equipped with sentience, able to consciously feel pain and a depressive mood when it is run for too long without a break. Does the sentient robot have moral standing? It appears so. If it can feel pain and depression, then it would be cruel to let it clean your apartment for too long, without a reason. Again, we have two cases: a robot vacuum cleaner with and without sentience. Since the presence of sentience is the only other difference between these cases, the difference in moral standing must be explained in terms of sentience.

20Someone might deny that the robot would have moral standing, even if it was sentient. However, if one seriously tries to empathize with the robot and its pain, this seems implausible. In light of the previous argument that the presence of valenced experiences always provides us with non-instrumental moral reasons, another possibility is more likely. If we deny that robots would have moral standing, even if they were sentient, then we are biased against attributing moral standing to artificial systems. This is first and foremost because it is inconvenient for us: attributing moral standing to beings means that we

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have to consider their interests in our decisions and that sometimes their interests may outweigh ours. With human babies or cute animals like the ones humans use as pets, we are emotionally drawn to empathize with and care about them. However, most AI systems don't have this emotional appeal to us, which is why our self-serving biases may get the better of us. In this way, our judgements about AI moral standing may be influenced by sources similar to biases preventing us from taking the possibility of insect moral standing seriously (Mikhalevich and Powell 2020).

I have put forward two arguments for weak sentientism. The view conforms to a meta-criterion according to which a being has moral standing if it has interests that matter to it, and it is supported by the method of cases. These arguments seem quite strong to me. At the same time, those are not knockdown arguments. It is perfectly possible to reject the proposed meta-criterion along with the principles and reasons I have appealed to in its defense. Similarly, if someone evaluates the cases I mentioned, and similar ones, categorically different from me, it is hard to see how these disagreements can be reconciled. At some point, I cannot do much more than hope that many readers share some of my foundational inclinations and intuitions. There is no deeper evidence I can point to.

Unfortunately, this predicament is not mine alone. In value theory and debates on moral standing, there are no agreed upon sources of compelling evidence.[7] Considering a variety of cases and plausible theoretical principles is the best that anyone can do. For this reason, I hold two positions which might seem in tension: First, the arguments in favor of weak sentientism are stronger than the arguments for any similarly substantive view in this research area. Second, it is still unclear whether weak sentientism is actually true. However, in our current state of knowledge, we should have a higher confidence in this view than in any other view on the grounds of moral standing, and typically act as if it is true.

In the next sections, I will defend weak sentientism against two of the most important objections to it.

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2.3 Defending weak sentientism I: The introspective objection

Let us begin with the *introspective objection*, proposed by Kammerer (2022).[8] It has the following form:

P1

Reductive physicalism is true.

P2

Reductive physicalism entails that beliefs about sentience which are claimed to be justified mainly by introspection are actually unjustified.

P3

Our belief that sentience grounds moral standing is claimed to be justified mainly by introspection.

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Our belief that sentience grounds moral standing is unjustified.

Reductive physicalism is the view that consciousness is reducible to, i.e., nothing over and above, physical states. Without elaborating on this debate (Bayne 2021, chap. 2; Rolffs 2023, sect. 2.3), one may take physical states to be the general kind of states described by natural science, particularly fundamental physics. So, physicalists say that consciousness can be completely accounted for by the properties or concepts referred to in the sciences, particularly physics.[9] In addition, physicalists also believe in functional properties, which are properties characterized by their causal role which is itself taken to be explainable in objective scientific terms. For example, a doorstop is characterized functionally as whatever prevents a door from closing. There is a perfectly scientifically intelligible explanation of how the doorstop performs this function which appeals to its weight and its position, relative to the door.

Introspection is the process by which one represents one's own mental states, in a manner which leads to self-knowledge and is first-personal (Franco 2024). For example, one way of learning what I really feel about something is to introspect, to "look inside myself". This contrasts with more intersubjective, non-introspective knowledge, such as knowledge acquired by means of seeing, hearing, or logical reasoning.

Kammerer assumes that our introspection depicts consciousness as a special kind of state. When we turn our attention (inward) to our conscious experiences, they appear as intrinsic (existing independently of other things), immediately known (with certainty), and irreducible to physical or functional properties (Chalmers 1996).[10] All these putative properties of consciousness are typically denied by reductive physicalists. Most clearly, by definition, reductive physicalists cannot grant that consciousness is irreducible.

22Given this, if reductive physicalism is true, introspection shows consciousness differently than it really is, i.e., it is illusory. Now, enter the following assumption: If our introspective grasp of consciousness heavily misleads us in this case, then we cannot trust introspection in further cases. This assumption has at least some merit: If we find out that a putative source of knowledge has misled us, and we have no strong evidence that it is otherwise reliable, it is rational to stop relying on this source. This would entail P2; i.e., the claim that beliefs about sentience which seemed to be justified mainly via introspection are not actually justified.

The question raised by P3 is: Do the arguments in favor of weak sentientism, i.e., the view that sentience is a sufficient (but maybe not necessary) ground of moral standing, rest mainly on introspection? According to Kammerer, the plausibility of weak sentientism is based first and foremost on our immediate introspection of the goodness or badness of our conscious experiences. He says:

> There appears to be no need for careful reflection on elaborated moral laws, ethical principles, or contractual agreements, to know that *this* (focusing on our current pain) is bad because of the way it *feels*, and hence that, when someone else feels the same thing, it is also bad.

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Kammerer concedes that there may be non-introspective justifications of weak sentientism. However, even so, the overall justification of weak sentientism would be weakened substantially once introspective justifications are undermined. He states (Kammerer 2022)[11]:

> Our impression that phenomenal consciousness has some significant intrinsic value at best becomes on a par with other impressions we might have about other valuable things, which means that we should treat it with great caution and consider it highly defeasible (think about Greeks and slavery, Kant and masturbation, etc).

That is, if we cannot rely on an introspective justification for weak sentientism, we should treat it as on a par with other questionable ethical views. There may be some highly defeasible reasons in favor of weak sentientism and it may ultimately turn out to be correct, but – according to the introspective objection – its epistemic status is not much different from culturally influenced prejudices. Thus, the introspective objection does not conclude that weak sentientism is false, merely that it is unjustified. Nevertheless, if Kammerer is correct and sentientism's epistemic standing is comparable to the justification beliefs like the permissibility of slavery enjoyed, weak sentientism would be in deep trouble.

23The question is: Does the introspective objection undermine the justification for weak sentientism we have provided? To avoid engaging in a debate which would deserve its own book, I grant P1 here, i. e., the truth of reductive physicalism. I have doubts about the other two premises of Kammerer's argument. An initial observation is that we have not explicitly adduced introspection as an argument in favor of weak sentientism. Instead, our argument was based on evaluating particular cases and the meta-criterion defended by Kagan and Ladak. However, one may claim that our argument still implicitly invokes introspection. According to this view, we introspect consciousness as valuable, and this process influences our evaluation of cases and whether we think that valenced conscious experiences generate interests that matter. For instance, when we think of someone in excruciating pain, we might imagine how it would be for ourselves to be in pain. Based on past introspection, we conclude that this would be very bad for us and infer that pain generates moral reasons to help the person in pain.[12]

A worry about this argument is that it may generalize to all proposed grounds of moral standing. As we will discuss in Chapter 5, some researchers are sympathetic to the view that having desires is a sufficient ground of moral standing. This includes Kammerer and Kagan (2019). Partially, this view is also based on considering cases. Suppose a being is not conscious but has a strong desire to, e.g., play football and act in accordance with this desire so that it seeks out opportunities to play football, expresses profound disappointment when prevented from playing, etc. It may seem immoral (for the being's own sake), in the absence of a good reason, to prevent this being from playing football. If so, then the being has moral standing.

However, introspection may be involved in these case judgments as well. When evaluating whether we have a moral reason to satisfy the being's desire, we might – in accordance with the worry above – ask ourselves whether it would be bad for us to have such a desire frustrated. In our case, this would typically involve a negative conscious experience, and this might be why we judge desire frustration as

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bad. Since we cannot trust our introspection of conscious experience, according to the introspective objection, we would be unjustified in believing that desires ground moral standing, if such intuition may be caused by introspection of conscious experience.

The same objection might be, maybe with varying grades of attractiveness, made to all putative grounds of moral standing. For instance, Taylor (1986) argues that all living beings, including plants, have moral standing because they all have a good of their own which they strive toward, namely, the full development of their biological powers. The introspective objection 24to this view would be: When considering how a plant or a bacterium strives toward its own good, e.g., survival and reproduction, we might imagine how it feels for us to strive toward the development of our biological possibilities. In us, this is typically associated with conscious experiences which makes it seem valuable. However, in plants or bacteria, consciousness is likely absent, and, furthermore, we cannot trust our introspective impression that consciousness is valuable.

To put my counterargument to the introspective objection more generally: According to the objection, our judgements about cases and theoretical meta-criteria might be indirectly influenced by introspection of conscious experience and thus unreliable. However, if this is accepted, then there arises the threat that arguments for grounds of moral standing other than sentience also invoke intuitions influenced by introspection of conscious experience. This is doubly problematic: First, the introspective objection entails that we cannot trust these intuitions. Second, if we could trust them, these intuitions would support the view that consciousness or sentience grounds moral standing, rather than whatever alternative ground of moral standing was advocated for.

If the introspective objection generalizes to all other putative grounds of moral standing, then its soundness would entail that all claims according to which something grounds moral standing are unjustified. I cannot rule out that this is the case. However, in this book, I work based on the assumption that we can make some tentative assessments of the foundational ethical issues related to AI suffering. That is, I rule out that a version of moral skepticism, according to which there is a wide domain of interest which does not admit of moral knowledge, is true.[13]

I also have a more direct counterargument to the introspective objection, however. This counterargument targets P2, i.e., the premise that reductive physicalism undermines the evidential value of introspection when it comes to beliefs about consciousness. The introspective objection is based on the assumption that introspection of conscious states is unreliable. However, things are not that clear-cut. According to the physicalist views at issue, introspection portrays consciousness as intrinsic, immediately known, and irreducible to physical or functional properties: all properties which consciousness is denied having. However, reductive physicalists do grant that humans know things like when they are in pain, what they are consciously seeing, etc. If people have this knowledge, they must have it via introspection. So there seem to be “good” and “bad” types of cases: Types of cases where introspection systematically deceives us and types where it is very reliable. To find out whether introspection of normatively significant properties belongs to the good or the bad type, we have to ascertain what members of the bad type have in common.

25It seems like all members of the bad type encapsulate a common core of dualist intuitions, i.e., intuitions according to which consciousness is a unique, metaphysically special property. In showing consciousness as intrinsic, immediately knowable, and irreducible, introspection depicts consciousness

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as something which is not an ordinary functional or physical property. By contrast, introspectively acquired beliefs like “I am in pain” and “I am currently seeing a red tomato” are metaphysically neutral. They are not in tension with the belief that consciousness is a physical or a functional property.

What about the introspectively acquired belief that sentience grounds moral standing? Prima facie, this belief is compatible with holding that consciousness is reducible to physical properties. For there is no contradiction involved in thinking that the same property is purely physical and normatively significant. However, even if the claim that sentience grounds moral standing is compatible with different accounts of its metaphysical nature, a proponent of the introspective objection may hold that the normative relevance of sentience is not justified without anti-physicalism. She may elaborate as follows: We only have the intuition that conscious pain is normatively significant because our introspection conceives of it as a non-functional process. If we regard consciousness as a purely functional process, e.g., a process of making contents globally available to different systems in the brain (Dehaene 2014; Mashour et al. 2020), why would it be normatively relevant? There is nothing about sentience, conceived in third-person terms, which seems normatively relevant. Thus, introspection of consciousness as normatively relevant, and its justificatory role, does stem from introspecting it as irreducible.

I disagree. First, this argument threatens to prove too much. If you think that all properties of humans are ultimately physical, then you must say that the grounds of moral standing, whatever they turn out to be, consist ultimately in physical properties. It is not clear that the functional or neuroscientific features underlying sentience are less plausible candidates for being grounds of moral standing than the scientific (e.g., neuroscientific) properties underlying desire, agency, or some other proposed ground of moral standing. That is why you should be open to the view that the physical properties underlying sentience indeed suffice for moral standing.

Second, I submit that the justificatory role introspection has for the normative relevance of consciousness does not depend on the problematic kind of introspection which portrays consciousness as non-physical. Introspection of consciousness as normatively significant does not depend on the introspection of consciousness as non-physical. As Kammerer (2022) explains himself, when introspecting conscious pain, for instance, we seem to recognize that *this* (kind of state) is bad because it *feels* bad. 26Similarly, Muehlhauser (2017) argues that our intuitions about the value of experience stem from demonstrative judgements. According to him, when my foot hurts (say), my intuitive judgement about the pain is roughly of the form “Whatever *this* is, *this* is really bad”.

So, our introspection picks out the value of consciousness demonstratively, just as the value of *this* property, while leaving it open what *this* is. If so, then our introspective judgements do not conceptualize consciousness in any theoretically laden way. It follows that our introspection is neutral regarding the metaphysical status of consciousness. In particular, it is neutral between whether consciousness is a physical property or not. Thus, introspection of consciousness as normatively relevant does not depend on anti-physicalist introspection. For this reason, physicalist views do not undermine the justification of the view that sentience is normatively relevant. The introspective objection fails.

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2.4 Defending weak sentientism II: The epistemic objection

The second objection to (weak) sentientism we will refute is the epistemic objection. In a nutshell, the epistemic objection runs as follows (Dung 2022b):

P1
In the foreseeable future, for many kinds of beings, there is no scientific consensus on whether they are sentient.

P2
We need to make attributions of moral standing now, for various practical purposes.

C
We should choose a criterion of moral standing which is not based on sentience.

The reasons in favor of P1 will become clearer in the next chapter. Here, it suffices to note that there is no consensus on which animals and AI systems are sentient. According to one extreme, consciousness (and thus sentience) is restricted to humans or only language-using humans (Carruthers 1989, 1992). According to another extreme, all complex systems with recurrent processing or even literally all material entities, including elementary particles (Goff 2017), are conscious. All kinds of intermediate views are represented in the literature (Carruthers 2020; Key 2016; Klein and Barron 2016; Le Doux 2019; Segundo-Ortin and Calvo 2023). Moreover, there are deep metaphysical (Schwitzgebel 2020) and methodological (Irvine 2012) controversies, so a resolution of these disagreements in the near future is very unlikely.

At the same time, as P2 states, the need for an account of the distribution of moral standing is pressing. There are many animals which may be 27subject to grave harm and injustice in case they have moral standing. Furthermore, without an account of moral standing, we don't know how close we are to designing AI systems toward which we have moral obligations. If we cannot reliably detect sentience, then we cannot know which beings have moral standing. Thus, we cannot use sentience as a criterion for moral standing in practice. Given that there is an urgent practical need for this knowledge, we need to resort to a different criterion of moral standing for the time being.[14] Then, proponents of the epistemic objection advocate for a criterion of moral standing which is claimed to be more amenable to empirical investigation, such as: behavioral (Danaher 2020) or cognitive equivalence (Shevlin 2021) with paradigmatic possessors of moral standing or objectively observable relationship to humans, including human responses to and interactions with candidates for having moral standing (Gunkel 2018, 2019).

So, the epistemic objection uses the lack of knowledge (and knowability) of sentience as a reason to reject sentience as the criterion of moral standing. However, the objection is misguided. To see this, we have to clarify a distinction implicit in the argument: the distinction between a *ground* and a *criterion* of moral standing. Weak sentientism is the claim that sentience is a sufficient ground of moral standing.

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That is, if a being is sentient, then it has moral standing *in virtue of* being sentient. Weak sentientism is thus a metaphysical claim which is motivated, criticized, and defended via the methods proprietary to normative ethics.

By contrast, a criterion of moral standing is used in practice to ascribe moral standing. For this reason, such a criterion is sensitive to pragmatic and epistemic factors. It must – given actual constraints of time and other resources – be possible to ascertain whether a being satisfies the criterion or not. Otherwise, the criterion is useless. Phrased in terms of disciplinary boundaries, questions about the criterion of moral standing belong to applied ethics.

To further illustrate the distinction: H_2O is the ground of water, i.e., an object is water in virtue of having the chemical structure H_2O. Being transparent, tasteless, odorless, and liquid at normal room temperatures are among the criteria of water; i.e., these are the features we often use in practice to identify whether something is water.

So, there is an immediate tension: The epistemic objection is most naturally expressed with respect to the criterion of moral standing, while (weak) sentientism is a claim about the grounds of moral standing. One could be tempted to conclude that the epistemic objection and weak sentientism are orthogonal: both can be legitimate. Yet this would be a mistake. To challenge weak sentientism, the epistemic objection needs to support at least one of two theses: First, it may suggest that weak sentientism is false; i.e., 28sentience is not a sufficient ground of moral standing. If weak sentientism is false, then – in the absence of further argument – we have no reason to use sentience as the criterion of moral standing. Second, proponents of the epistemic objection may hold that weak sentientism is true, but that sentience is nevertheless not the criterion of moral standing. In other words, sentience may be the (sufficient) ground but not the criterion of moral standing.[15] In this case, weak sentientism's practical significance would be in doubt.

I will argue now that the epistemic objection supports neither of these two claims. If so, it is irrelevant to sentientism. Let us first discuss the first type of objection. That is, does the epistemic objection show that weak sentientism is false? It seems clear that it does not. Weak sentientism is a metaphysical claim about the grounds of moral standing. Its truth depends on what moral standing ultimately consists in. Whether or not weak sentientism is true is examined with methods of normative ethics. Paradigmatic methodologies are the ones we relied on earlier: considerations of particular cases and plausible theoretical principles. The outcome of these procedures is independent from the epistemic accessibility of sentience.

More generally, the claim that sentience grounds moral standing does not imply, or even suggest, that we can reliably test for sentience in most beings. For it is perfectly coherent, although it would be unfortunate, that facts about moral standing are unknowable to us. To illustrate this, consider that water might be said to be grounded in its chemical structure. Even if most objects – including instances of water – could not be tested for their chemical structure, this does not refute the contention that water is determined by H_2O. Analogously, even if the distribution of sentience cannot be known, this is compatible with the claim that sentience determines moral standing.[16] To conclude the rejection of the first interpretation of the epistemic objection: Since (weak) sentientism is consistent with the

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unknowability of sentience in non-human animals and machines, the epistemic objection does not threaten (weak) sentientism.

Let's move on to the second interpretation of the epistemic objection. Does the epistemic objection provide reason to believe that, even if (weak) sentientism is true, sentience is not the criterion of moral standing? It does not. Something can only be a criterion of moral standing if it correlates with a ground of moral standing. Consider a dilemma.

On this interpretation of the epistemic objection, we assume that sentience is sufficient for moral standing. Suppose someone now proposes a criterion of moral standing according to which the detection of property F should be taken as providing sufficient warrant for ascribing moral standing. To overcome the epistemic objection against sentience as a criterion, F needs to be a property that we can reliably detect. Tautologically, F either 29reliably tracks the presence of sentience or does not reliably track the presence of sentience. If F reliably tracks sentience, then F can be used as a measure of sentience. In this case, the epistemic objection rests on the false premise that sentience cannot be measured. By relying on F as an indicator of sentience, we can measure the distribution of sentience and thus assign moral standing accordingly. Given this horn of the dilemma, F serves as a criterion of moral standing by being an indicator of sentience, contrary to the conclusion of the epistemic objection.

According to the other horn of the dilemma, F does not reliably track sentience. If F does not reliably track sentience but sentience is the necessary and sufficient ground of moral standing, then it follows that F does not track moral standing.

By joining the conclusions of both horns of the dilemma, we see that any feature F either serves as a criterion of moral standing by being a reliable indicator of a ground of moral standing or does not track moral standing and thus cannot be used as a criterion for it. Granted, weak sentientism allows that there are grounds of moral standing other than sentience that F may track. So, F may track moral standing in virtue of tracking grounds of moral standing other than sentience. However, this does not change the previous lesson. Any feature F needs to either be an indicator of some ground of moral standing or cannot be used as a criterion of moral standing. Moreover, crucially, epistemic considerations are irrelevant to whether something is a ground of moral standing.

We can conclude that, given weak sentientism, any putative criterion of moral standing either is an indicator of sentience, or of another sufficient ground of moral standing, or fails since it does not in fact indicate moral standing.

The water analogy illustrates this inseparability of ground and criteria of moral standing. Given that the chemical structure H_2O grounds the property of being water, superficial features of water like its lack of taste, odor, and color can indicate the presence of water – and thus serve as its criterion – only insofar as they indicate the presence of H_2O. It would make no sense to propose an alternative criterion of water which does not presuppose our ability to detect H_2O, if we grant that being H_2O is necessary and sufficient for being water. However, this is structurally analogous to what proponents of the epistemic objection do if they grant that sentience grounds moral standing.

We have arrived at the following results: If sentience is a necessary and sufficient ground of moral standing, then an investigation of the distribution of moral standing has to be an investigation of the distribution of sentience. Since epistemic considerations are irrelevant to whether sentience grounds

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moral standing, they are irrelevant to whether sentience is 30a proper criterion of moral standing. If sentience is sufficient, but not necessary for moral standing, then it could be wise to focus more on grounds of moral standing which are epistemically more accessible than sentience. However, on most accounts, alternative grounds of moral standing will not always be present if and only if sentience is present.

If so, we will encounter beings which have moral standing, if they are sentient, but lack moral standing if they are not sentient. In these cases, we do not have any choice but to make assessments of whether beings are sentient which are as informed by scientific evidence as possible. It is not justified to switch to independent criteria of moral standing just because sentience is hard to discern. For if these criteria do not track sentience, they are not criteria of moral standing. This is why the epistemic objection fails. The ethical relevance of sentience is independent of its epistemic accessibility, or its lack thereof.

This conclusion conforms to the general observation that we should not abstract from ethically significant facts in our moral decision-making even if they are very hard to detect. For example, imagine there is suspicion that someone committed a murder. Imagine that it is very hard to find out whether the person is guilty. Nevertheless, whether this person should be imprisoned clearly depends on whether they committed the murder. There is no use in coming up with independent, more epistemically accessible criteria for the decision whether to imprison them. In the absence of knowledge whether the person is guilty of the murder, we can only use our best judgment under uncertainty: based on how likely it is that the person committed the murder and on what the best course of action is, given this probability. For example, legal systems typically err on the side of presuming defendants to be innocent, unless the opposite is highly likely.

In this chapter, I have explained weak sentientism and related views and concepts, have provided two arguments for weak sentientism, and have refuted two important objections to it. In the next chapter, we will examine how to detect empirically whether and which AI systems are sentient.

Notes

1. This excludes the special case, discussed also in this chapter, of sentient creatures which can only have positive experiences. Return to text.#
2. The latter view is entailed by “hedonism” about wellbeing (Bramble 2016; Crisp 2006; Moore 2019). Return to text.#
3. Moreover, suppose – for some reason – someone denies the soundness of the subsequent arguments for the connection between either sentience and moral standing or sentience and welfare subjecthood, but not both. In this case, one could argue for all the claims I make here by using one of these arguments and supporting the independent principle that moral standing and welfare subjecthood are equivalent.
   31In fact, denying the equivalence between welfare subjecthood and moral standing can even strengthen the case for believing that AI systems will soon matter morally. Mogensen (2025) argues that being a welfare subject implies having moral standing but that moral standing (in

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virtue of having autonomy) can exist without welfare subjecthood. This strengthens the case for believing that AI has moral standing, as it provides a further (possibly independent) route how AI systems might have moral standing.
Return to text.#

4. "Pro tanto" reasons speak in favor or against actions but may be outweighed by other reasons. For example, that being high is fun provides a pro tanto reason to consume many kinds of drugs, but I may, all-things-considered, not have sufficient reason to consume them (because of health risks, my other responsibilities, etc.). When I speak of reasons in what follows, these are usually pro tanto reasons. For simplicity, I will mostly omit the pro tanto qualifier in what follows. Return to text.#
5. This principle overlaps significantly with the well-known resonance constraint on wellbeing, according to which things can only be non-instrumentally good for a person if they resonate with this person or are suitably connected to what the person finds attractive (Fanciullo 2025; Railton 1986). Return to text.#
6. I will discuss this issue in more detail in Chapter 5, where we will encounter the question whether the facts determining a being's wellbeing also determine whether a being is a welfare subject in the first place. Return to text.#
7. There are methodological approaches which aim to go beyond this methodological problem, but they fall short. For reasons of space, I can only refer to experimental philosophy (Machery 2017) here. In my view, this approach cannot satisfactorily explain why laypeople's intuitions on cases should have philosophical significance (Deutsch 2010, 2015). Return to text.#
8. For a further argument against sentientism, see Kammerer's (2022) indeterminacy argument and related arguments by others (Birch 2022; Cutter 2017; Lee 2019; Pautz 2017). For further discussion, see my counterargument (Dung 2024) and Kammerer's (2024) reply. Return to text.#
9. There is a variety of physicalist views on the market. Strong illusionism (Dennett 1988; Frankish 2016) and type-A physicalism (Chalmers 2003) are reductive physicalist views which claim that consciousness is fully reducible to physical states (or they can be interpreted as eliminativist views which say that consciousness does not exist). Type-B physicalists (Balog 2012; Diaz-Leon 2016; Papineau 2002; Tye 2008) believe that consciousness is metaphysically but not epistemically reducible to physical states. In other words, they hold that consciousness is nothing over and above physical states, but that our *knowledge* about consciousness is irreducible to our knowledge about physical states. These differences are not central here. Return to text.#
10. While I grant this assumption here, it is not obvious. According to some accounts of the meta-problem of consciousness (Chalmers 2018), i.e., the problem of explaining why it seems like there is a special and unique problem of explaining consciousness in physical terms, the intuition that consciousness is intrinsic, immediately known, and irreducible is not caused by introspection, but by contingent cultural and sociohistorical factors (Rosenthal 2019). If so, then one may argue that there are no analogous undermining factors present in respect to our intuition that consciousness grounds moral standing. Thus, one may say that we can trust this latter intuition more. Return to text.#

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11. Kammerer speaks here about intrinsic value, not moral standing, but the relevant arguments are also meant to apply to moral standing. Kammerer targets his arguments at sentientism in general, not specifically weak sentientism. Yet they straightforwardly apply to weak sentientism as well. Return to text.#
12. 32This is not the only way how our case judgements could be 'infected' by introspection. For example, it could also be that introspection causes a standing belief that consciousness is intrinsically valuable and that this belief influences our evaluation of particular cases. Return to text.#
13. That being said, this knowledge may be fallible and uncertain. Return to text.#
14. See Dung (2022b) for an overview of instances of the epistemic objection in the literature. Return to text.#
15. This view is quite explicitly endorsed by Danaher (2020). Return to text.#
16. The analogy even oversells the relevance of the knowability of sentience to the justification of sentientism. In the case of water, we need to test the chemical structure of some samples to infer that water is grounded in H_2O. In the case of sentientism, we mainly do not proceed via identifying specific possessors of moral standing and examining whether they are sentient. Instead, we use, e.g., the method based on case judgements sketched previously. Return to text.#

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5

NON-CONSCIOUS SUFFERING AND AI

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5.1 Consciousness without sentience and non-conscious suffering

In Chapter 2, I have argued that weak sentientism is true: Sentience is a sufficient ground of moral standing and the capacity for suffering. While this thesis rules out that some sentient beings lack moral standing, it leaves open whether some non-sentient beings have moral standing. In this chapter, I investigate which properties besides sentience may ground the capacity for suffering.

This section briefly considers whether consciousness without sentience may ground the capacity for suffering. The next section investigates which form of agency is a suitable candidate for grounding the capacity for suffering. Then, the subsequent section examines whether theories of wellbeing provide evidence for the possibility of non-conscious suffering. My conclusion is that a plausible case for the existence of grounds of suffering independent from sentience exists, but that the issue remains very much open. The fourth section of this chapter assesses whether near-future AI systems likely possess these potential grounds of non-conscious suffering. Then, the last section concludes my argument for near-future AI suffering risk.

I take it to be obvious that two things hold:

a. Our credence that properties other than sentience are sufficient grounds of the capacity for suffering should be above zero.
b. Our credence that near-future AI systems have these properties should be above zero.

104In the previous chapter, we have only considered the possibility of AI sentience. If (a) and (b) hold, then we should increase our degree of belief that near-future AI is capable of suffering, relative to an assessment which only considers AI sentience.

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This chapter investigates whether our credence in sufficient grounds of suffering other than conscious valenced experience should be high and whether the chance that some near-future AI systems have the relevant properties is high. If so, this should significantly increase our credence in near-future AI suffering. I will begin with the first question: Why should we think that properties other than sentience ground moral standing?

Before presenting the arguments, I will set aside the view that conscious experience which is not valenced can ground moral standing. In Chapter 3, we have already seen weaknesses of this view. The assessments of particular cases (e.g., the Vulcan) which favor this view are either very contestable or can, with some plausibility, be regarded as support for the view that some non-conscious properties ground moral standing. That being said, I am open to the view that consciousness, even if not valenced, suffices for moral standing.

However, it seems clear that, all other things being equal, the experiences of beings which are merely subject to non-valenced conscious experiences matter a lot less than of beings which can have valenced conscious experiences like pain or fear. For, in the case of humans and other animals, the kinds of experiences which generate the strongest moral claims for help are precisely the ones which feel particularly unpleasant, such as the experiences caused by torture. Hence, for practical purposes, the question whether machines have non-valenced consciousness is less of a priority than questions about sentience and some other potential grounds of non-conscious suffering in AI.

There is an exception to the claim that non-valenced experiences matter much less (all other things being equal) than valenced consciousness. According to Roelofs (2022), a being has moral standing if it has the capacity for motivating conscious states. These are "conscious states which participate on the conative side, by making some option, outcome, or action appear good or attractive" (Roelofs 2022). Suppose that there are non-valenced conscious experiences which are nevertheless motivating. In virtue of their connection to conative states, i.e., states connected to what is traditionally called the "will", such motivating conscious experience may matter ethically to a significant extent. Perhaps, mistreating beings with strongly motivating non-valenced conscious states is bad, to a similar degree as causing strongly negatively valenced experience is bad. For this reason, it seems like the truth of Roelofs' view would make potentially a big difference to how we should treat AI systems.

105That being said, in practical circumstances – which are dominated by empirical uncertainty regarding the mental properties of AI – our evaluation of this view does not change our assessment of the risk of AI suffering. First, as Roelofs notes, it may be the case that only valenced conscious experiences are motivating in the relevant sense. In this case, Roelofs' view would have the same implications as strong sentientism. Second, it is currently not possible to empirically distinguish the presence of sentience, motivating consciousness and agency, and desires in AI. In other words: We don't know of features of AI which would be evidence of sentience, agency, or desires, but not of motivating consciousness, or of motivating consciousness but not of sentience, agency, or desires.[1] Hence, in Section 5.2, we will not explicitly discuss indicators of motivating consciousness. However, the indicators we will discuss can also be treated as partial indicators of motivating consciousness.

Obviously, views which tie moral standing to (motivating) consciousness may be more restrictive than the views we will discuss. This is because they may claim that *only* conscious beings have moral

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standing. Understood in this way, the subsequent arguments aim to put views according to which consciousness is necessary for moral standing into doubt. Ultimately, I am uncertain myself about whether moral standing and suffering require consciousness or even sentience. However, there plausibly are cases where not treating AI systems in certain ways has negligible costs to humans but may save AI systems from significant suffering if they are capable of suffering. Given this uncertainty – or so I will argue – we should extend some moral consideration to AI systems which are not sentient but possess other properties which may well constitute suffering.

I think there are at least three important sources of arguments for the view that non-sentient beings might have moral standing which need to be considered.[2] First, the sources we already encountered in Chapter 2 when making the case for weak sentientism: theoretical principles and consideration of various cases. The third is discussed in Section 5.3: theories of wellbeing.

5.2 Non-conscious suffering and agency

One of the main competitors to sentientism is the view that agency grounds moral standing. Since we already ruled in favor of weak sentientism in Chapter 2, we will only scrutinize the view that agency is a sufficient, but not necessary, ground of moral standing. So, we will first clarify what agency is and then discuss whether theoretical principles and evaluations of particular cases suggest that agency suffices for sentience.

What is agency? Answering this question is far from easy. Many different accounts and characterizations of agency can be found in the literature. 106In particular, there are two challenges associated with singling out a proper characterization of agency. First, it is ambiguous to what extent different characterizations embody conflicts between different *accounts* of agency or between different *concepts* of agency. If we are confronted with various accounts of agency, then agency is to be understood as their shared target phenomenon, and there is a substantive question about which account best describes and explains it. If there are varied concepts of agency, then our task is to agree on which concept, or concepts, of agency to use in which contexts. There may not be a substantive question about which concept is best or which one is *really* agency. Different concepts can pick out different legitimate properties. Which of these to call “agency” may be merely a matter of convention, rather than a philosophically very deep question.[3]

Second, some characterizations of agency are much more demanding than others, which raises problems when discussing whether agency grounds moral standing: For very restrictive conceptions of agency, it will be much more plausible that they ground moral standing than for very permissive conceptions.

In this chapter, we do not aim to settle the question of what the correct account of the metaphysical nature of agency is. Yet we also do not engage with the question of what the most useful concept of agency, all things considered, is. Instead, we only try to extract, from the literature, a concept of agency which is relevant to debates on moral standing. Hence, it should be a concept which is possibly a

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sufficient ground of moral standing and does not entail the presence of sentience, since a concept of agency which implies sentience would have no relevance to the distribution of moral standing (given weak sentientism).

To get a first grip on the notion of agency, we should note that every conception of agency involves the capacity to pursue goals *in some sense*. The literature in different disciplines – philosophy, psychology, and AI – agrees that agents are goal-directed (e.g., Glock 2019; Kenton et al. 2022; Liljeholm 2021; Orseau et al. 2018). This is a pretty weak constraint on agency.

We have already characterized a minimal conception of goal-directedness in the last chapter. According to a minimal conception of goal possession, an agent pursues goals if there are certain world states (or states of its own body) which it seeks out in various situations and which it tries to maintain in the face of perturbation. This amounts to a conception of agency which applies to a similar or the same set of things as Dennett's interpretivism (Dennett 1987, 1991). According to Dennett's view, a system has goals when interpreting it as goal-directed has value for predicting or explaining its behavior. This view entails that, e.g., a chess-playing system may have the goal "capture the opponent's queen" since ascribing this 107goal helps to make sense of their behavior. In addition, even bacteria reliably avoid damaging stimuli, so they can in a minimal sense be characterized as pursuing goals related to maintaining organismic integrity. Since it seems unlikely that we should ascribe moral standing to bacteria or to every functional chess computer, we need a more demanding conception of agency for our purposes.[4]

More demanding forms of goal-directedness are suggested by Butlin (2020, 2022, 2023). Following Butlin (2020) and a wider literature in psychology (Dickinson and Balleine 2000), we can distinguish so-called goal-directed and habitual control of behavior. In goal-directed control, systems employ independent representations of the value of outcomes and of dependencies between actions and outcomes to calculate the expected value of possible actions (Butlin 2020). The expected value of an action is determined by the values and the likelihoods of each possible outcome of this action. By contrast, in habitual control, values of actions themselves are learned from experiences and used in action selection. The system does not attach values to all potential outcomes to calculate which actions are most likely to lead to desirable outcomes. Instead, it learns directly from past actions, which actions tend to lead to good outcomes. So, habitual control does not require an explicit representation of the possible states of the world and how they are causally related.[5]

Perhaps we can say that the relevant sense of agency is limited to goal-directed control. For while habitual control manifests preferences of beings between world states, their behavior is only indirectly sensitive to the value of possible outcomes of actions. These values do not figure explicitly in action selection. Only in goal-directed control is the behavior of beings directly sensitive to the value of possible states of the world and thus goal-directed in a more demanding sense. Furthermore, it is not clear how widespread goal-directed control is, for instance, in the animal kingdom.[6] While habitual control is very widespread, this opens up the possibility that the kinds of beings capable of goal-directed control might match well with the number of beings we would pre-theoretically be prepared to attribute moral standing to. Hence, the worry that agency as sufficient ground of moral standing is wildly over-permissive may not apply to a notion of agency as goal-directed control.

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Butlin (2022) claims that a being can only be an agent if it can learn to produce outputs for their instrumental value. That is, an agent must be able to learn that achieving a certain outcome might have value, not (only) because it is intrinsically sought, but because it makes it possible to later attain a goal state. According to Butlin, behaving in accordance with instrumental relationships can only count as an instance of agency if the behavior is learned, not if it is hard-wired. Only the ability to *learn* 108 instrumental relations which shape behavior is what makes systems able to flexibly pursue their goals and is thus conducive to agency. With few select exceptions, all reinforcement learning systems fulfill this criterion, but many other AI systems do not (Butlin 2022). Hence, one may be worried that, when learning to produce outputs for their instrumental value is seen as sufficient for agency, agency is an over-permissive ground of moral standing. One may react to this problem, however, by more modestly holding that this property is only necessary, not sufficient, for agency.

There is a further conception of agency one may take to be related to moral standing. According to the classical philosophical conception of agency, having agency means having the capacity for intentional action (Davidson 1963; Schlosser 2019). To be an intentional agent in this sense, one needs to be able to act for reasons. Butlin (2023) argues that model-based reinforcement learning systems, which computationally implement goal-directed control, count as acting for reasons. This is because they represent the state transition function, i.e., which states of the world will give rise to which other states (conditional on certain actions), as well as the value of different states, and use this information to decide which response to output. Therefore, "they represent and act on the basis of facts which count in favour of their actions, given their goals, and therefore […] act for reasons" (Butlin 2023).

However, intentional action is typically thought of in a more demanding way (Dung 2024). In the human case, the reasons for action are typically taken to be, or to be represented by, beliefs and desires. This is because reasons for action arguably need to be (based on) personal-level mental states which cause behavior. Such states, as the name says, can be ascribed to the whole person or being. A person believes something. By contrast, sub-personal states can be attributed to individual parts of a being. For example, pumping blood is most naturally described as something that a human's heart does – not the human itself. Behavior driven by sub-personal mental states should not count as intentional action. For example, virtually all bodily movement involves low-level motor representations in the brain: the brain states which encode detailed information about how to move one's body. However, humans cannot reflect nor consciously experience these representations. They also cannot take any other voluntary attitude toward them. Partly for this reason, they are most naturally regarded as states of the human brain, not of the entire human person.

I think there is a plausible case to be made that not even every system which has beliefs, desires, and similar states (called "propositional attitudes" by philosophers) that cause behavior has intentional agency in the sense relevant to moral standing.[7] This is because reasons may be restricted to mental states which are owned by the system, i.e., the mental states need 109to be, or represent, reasons *for* the system.[8] It seems plausible that further conditions are necessary for reasons to count as reasons *for* the system, or to be internal to it.

Such conditions may be found in the work of Frankfurt (1988). He argues that some thoughts (including desires) of persons can be "external" to them. Examples are intrusive thoughts (a possible

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symptom of obsessive-compulsive disorder, as well as other mental health conditions) and desires induced via hypnosis or drugs. Imagine the compulsive thought that you forgot to turn off your stove when you left home: you may remember turning off the stove, you may not want to think about the stove, and you may be sure that you turned it off. Nevertheless, the thought reoccurs again and again, without you being able to suppress it.

According to Frankfurt, such thoughts should not be attributed to the whole person; they are – in some sense – *external* to the person. Frankfurt provides several related justifications for this claim: the person is passive with respect to these thoughts, these thoughts may be in conflict with the person's self-conception (e.g., when they conceive of themselves as a person who does not succumb to irrational fears), and the person may experience themselves as "not in control of oneself" (e.g., when captured by anger which is experienced as an intrusion and which the person wants to vanish). Frankfurt expresses sympathy for the view that whether a thought is external to a person has something to do with the *higher-order attitude* a person takes toward it. At least, if a person approves of having a certain thought, then this thought is internal to them. It seems that a person can even make a thought or desire their own by endorsing it.

As described by Jaworska (2007a), not only Frankfurt but also many other influential researchers, such as Velleman (1992), Bratman (2000), and Korsgaard (1996), claim that internality requires the capacity to reflect on one's thoughts and desires. Systems can endorse their reasons in the right, reflective, way only if they can reflectively scrutinize them.

This view is not uncontested. Birch (2022) hypothesizes that reflective endorsement may not be necessary to make goals one's own; registering them in conscious experience may alternatively also suffice. Similarly, Jaworska (2007a, 2007b), motivated by the idea that a reflection-based view of internality cannot capture the moral weight of infants, argues that an emotion-based attitude of "caring" may be sufficient for internality. However, both authors are sympathetic to the view that reflective endorsement of reasons and goals is sufficient for moral standing, even if perhaps not necessary. Thus, such views are compatible with – and in support of – the view, considered here, on which reflective endorsement and sentience are both individually sufficient but not necessary conditions for moral standing.

110Thus, on such a view, a significant degree of metacognition would be necessary for agency. A being needs to be able to relate rationally, in the form of reasons-based and reasons-responsive reflection, toward its own goals. This is, excluding views where agency in addition requires consciousness, the most demanding plausible account of agency on offer. It is an open question whether there are other ways in which a system can come to own its reasons and thus be an intentional agent without such sophisticated cognitive capacities.

We have now assembled several conceptions of agency. On which conception, if any, is the view that agency is a sufficient ground for moral standing supported by plausible theoretical principles? Recall that, in Chapter 2, we have argued for a meta-criterion for ascertaining which beings have moral standing. According to Ladak and Kagan, beings have moral standing if and only if they have interests

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that matter to them. This meta-criterion supports the view that agency is sufficient for moral standing just in case the following holds: If a being has agency, then it has interests that matter to it. Call this conditional the *agency interest link*.

The agency interest link is more plausible on some conceptions of agency than on others. Take the view that agency equals goal-possession in the minimal or the interpretivist sense we outlined earlier. On this view, it suffices for being an agent that attributing agency to one is useful for prediction or explanation. Goal possession in this minimal sense seems clearly necessary for agency. However, it is too minimal to support a notion of agency that is linked in an interesting way to moral standing. For when we presuppose this notion of agency, the agency interest link does not obtain. Using this notion, we can say that agency implies the possession of interests, insofar as interests can be equated with minimal goals. However, the minimal notion of goal does not ensure that there is a subject to whom these goals matter. The goals may be the goals of a chess computer or a thermostat: We cannot say that goals matter to them.[9]

What about agency as the capacity for goal-directed control, or as the capacity to learn to produce outputs for their instrumental value? One can tell a story where these capacities presuppose that beings have goals, and therefore interests, in a more substantive sense of the term. For they involve a flexible sensitivity to the states the being prefers. Yet it is not clear why this would imply that the interests of such beings matter to them. Even if a system decides on actions based on the expected value of their outcomes, including their instrumental value, the system's goals might not matter to it.

Being capable of long-term planning, which presupposes sensitivity to the instrumental value of actions but is not entailed by it, plausibly affects the kinds of interests beings can have (Dung 2024). Beings with this capacity can have long-term goals and corresponding interests. These interests may 111 be relevant for the right of a system to not be killed. For, arguably, one thing that contributes to the harm of death is that it entails the frustration of long-term goals (Luper 2021; Singer 2011). However, again, whether interests are directed at the more distant future does not affect whether they matter to a being. So, what is relevant to whether interests matter to a being?[10]

It is plausible that a being's goals matter to it if they are *owned* or endorsed by the being. This suggests that a being is an agent in a relevant sense if it has mental states which cause behavior and are owned by the system, i.e., the mental states need to be, or represent, reasons and moreover reasons *for* the system. As we have seen, this could be the case if the being reflectively endorses its goals. If the being can reflect on these goals and judges that they fit its conception of itself and its more encompassing plans, desires, and ambitions, then a being owns its goals in the relevant sense.

Ways of owning goals other than reflectively endorsing them (and other than via sentience) may also be possible and – if so – might also suffice for having interests which matter to oneself. Arguably, goals could be owned by a being if they are integrated with all of its activities and reactions. If humans have an important goal, then they will often (for example) (a) seek out the goal in a wide variety of situations, (b) react with disappointment if their goal is frustrated, (c) have other goals which are conditional on the goal being fulfilled, (d) daydream about the fulfillment of their goal, and (e) regret if they fail to achieve their goal. Perhaps these kinds of activities and reactions, which integrate a goal with a being's wider life projects, behaviors, and emotions, suffice for goals to be owned by a being.[11]

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It is hard to evaluate whether satisfying this account of goal ownership is sufficient for moral standing. I have concerns that this account is too demanding, but also that it is too permissive. On the demandingness end, disappointment, daydreaming, and regret may presuppose the capacity for conscious experience. It is not clear whether we can ascribe these mental states, in the sense which matters to goal ownership, to beings which do not have conscious experiences. Even if non-conscious beings can be disappointed, daydream, or regret something, the intuition that these states might make the goals of beings matter morally might stem from conceiving of these states as conscious. If this route to goal ownership presupposes conscious experience, then it cannot motivate agency as a ground of moral standing which is independent from sentience.

On the permissiveness end, even plants or bacteria can be seen to have goals which guide their behavior in a variety of situations, and which are conditional on each other. These goals could concern survival and more concretely maintaining homeostasis, which is something all living beings strive toward. It follows that it is questionable whether goals that only have these two features suffice for moral standing.

112Hence, this suggests that, probably, only a notion of agency which involves the capacity to reflect on and consequently endorse goals suffices for having interests which matter to oneself while not requiring consciousness. That being said, one cannot yet confidently rule out that another notion of agency or goal possession which is also a plausible candidate for satisfying the agency interest link can be constructed.

However, it is plausible that agency requires some of the other capacities mentioned earlier in addition to reflection. If a being's other cognitive states were not integrated with the goal in question, it would be unclear whether the being has actually endorsed the goal. At a minimum, we should require that a being has the behavioral dispositions which come with having a goal, i.e., trying to achieve the goal across situations, reacting positively when the goal is fulfilled and negatively when it is not fulfilled, etc. If a being reflectively assented to a goal but did not have the behavioral dispositions that usually come with this, e.g., was not at all disposed to pursue the goal, it is unclear whether we should count it as having a goal.[12]

So, agency – as I understand it here – consists in multiple capacities. A being is an agent if it: (a) possesses goals which it robustly pursues across a range of situations, (b) behaves in ways indicating a negative evaluation of the situation when its goal is not fulfilled and a positive evaluation of the situation when its goal is fulfilled, and (c) reflectively endorses its goals. According to the previous argument, such beings count as agents, have interests that matter to them, and might thereby have moral standing. One may argue that the connection between agency and having interests that matter to oneself is even clearer than between the possession of such interests and sentience (described in Chapter 2). After all, having goals is almost synonymous with having interests. Having higher-order attitudes, reflective endorsements, about these goals provides a very natural way in which these interests can matter to oneself.

Besides, I am agnostic on whether condition (b) is really necessary for agency. The previous argument suggests that goal following behavior and reflective endorsement of goals might be jointly sufficient for agency. However, if the being shows more of the behavioral dispositions associated with goal possession in humans, as described by condition (b), the case for agency becomes more robust.

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The strongest counterargument against the view that all systems which satisfy conditions (a)–(c) are agents is: According to a certain conception of what it means to be a self or a subject, one can only be a subject if one has (unified) consciousness. Consciousness may be relevant here for two reasons: (i) It is connected to subjectivity in the sense of having a first-person perspective, a subjective point of view. (ii) Consciousness is, 113according to a common view, unified (Bayne and Chalmers 2003). All one's experiences are bound together in one conscious stream. Maybe this unity is required for being one subject.

If consciousness has these roles, then perhaps non-conscious agents could not have interests which matter to themselves because they are not selves? It is hard to settle this question. At the very least, there are possible conceptions of selfhood where selfhood only requires a sufficient degree of functional integration and perhaps embodied activity (Gallagher 2013; Newen 2018) or narrative structure (Dennett 1992), not consciousness. In general, one may have doubts that the notion of selfhood, which is implicit in the agency interest link, should be taken to be very demanding.

We will now look at thought-experiments involving non-sentient agents. This will, first, shed light on what such beings might be like and, second, allow us to test whether it seems plausible that the agents figuring in these cases have moral standing. The first relevant case is inspired by the Vulcan thought-experiment but does, of course, concern creatures not possessing consciousness. So, consider beings which have human-like intellect, perceive and act upon the world, and who lead their lives so as to achieve certain goals, e.g., create peace in the galaxy or help their friends. Yet those beings have neither affective experience, e.g., conscious pain, joy, and emotion, nor sensory phenomenology. While valenced experience registers violation or satisfaction of a being's interests non-reflectively (Birch 2022), these beings relate to their interests purely intellectually. They orient their behavior around their goals, respond to the frustration of their goals (e.g., by showing behavioral signs of disappointment), and regularly reflect on and revise or approve of their own goals (although these thought processes are not conscious). Call them "non-conscious paradigmatic agents".

It is hard to say whether we should judge that non-conscious paradigmatic agents have moral standing. Intuitions will diverge. There is a case to be made for ascribing moral standing to those agents. This can be brought out by thinking in more detail about how they behave. To achieve this, consider a more detailed thought-experiment developed by Gruen (2011) and also relied on by Kagan (2019): Suppose that extraterrestrials kidnapped you and are now planning to experiment on you, torture you, or something along these lines. You are trying to convince them to let you go back to earth without harm. According to Gruen, you

> might express your desire not to be held captive, that being held against your will is wrong, and that it prevents you from doing not just the things you want to be doing at home, but the things you are supposed to be doing. These aliens are frustrating your desires and preventing you from fulfilling your obligations to others. You might explain that you 114are a rational and sensitive individual who has immediate desires and long-term plans that you hope to satisfy. You don't think you should be treated as a means to some alien ends. You might try to bargain with them, telling them you will do something for them if they do something for you. You have relationships to others that you want to continue to pursue, and you would be

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> willing to develop a relationship with them if they respect you. You value your freedom and your ability to make choices. You need to be with your friends, family, and others of your kind. If you are forced to stay with the aliens, you will become bored, frustrated, lonely, angry, and depressed. You may even die. Holding you captive, against your will, harms you in many, many ways.
>
> *(Gruen 2011, p. 26)*

I take it that everyone should agree that it would be wrong for the aliens to not let you go back to earth if nothing else of high moral significance is gained by keeping you imprisoned. Moreover, it seems like the reasons given in this quotation are sufficient to make it wrong for the aliens to keep you captive. The big question is now: Could non-conscious paradigmatic agents *truthfully* adduce the same reasons against mistreating them as you can? If so, then non-conscious paradigmatic agents can be treated morally wrongly, which implies that they have moral standing.

Non-conscious paradigmatic agents have, by definition, desires and long-term plans, dispositions for rational and sensitive behavior, thoughts, the ability to bargain and plea for their lives, and relationships they value. It is unclear whether they can be bored, frustrated, lonely, angry, or depressed because they do not consciously feel. At a minimum, these states don't have the same significance as they have in humans. One may argue that desires, long-term plans, and valued relationships also don't have the same, or any, significance if they are not registered in conscious experience. It is hard to see how to refute, or confirm, this viewpoint.

That being said, this thought-experiment makes something salient: Imagine that we encounter a non-conscious paradigmatic agent, like the one in the thought-experiment, in the real world. It seems clear that we would be emotionally compelled to treat it as having moral standing (Shevlin n.d. makes a similar prediction). It would strive to achieve its goals, preserve itself, talk to us, and exercise rationality. We could not bear to unnecessarily harm such an agent, even if we think that it is not conscious.

Our moral intuitions would be partially sensitive to the appearance of this agent. If the agent looks more human-like, then the appearance that it has moral standing is stronger (Küster et al. 2021; Nijssen et al. 2019; Thellman et al. 2022). However, if the agent is sufficiently intelligent, including being 115 able to communicate with us, behave flexibly, and pursue long-term goals, then the agent will appear to have moral standing, even if it does not have a human-like body or face. Crucially, the strong intuition that the agent has moral standing will remain, it seems to me, even if we know that it is not conscious.

I think that it is a plausible conjecture that there are non-conscious paradigmatic agents which would intuitively appear to us to have moral standing if we encountered them. Moreover, I think that we should give some weight to this conjecture when judging whether such agents have moral standing. While one may dispute how these cases should be evaluated, there seem to be at least cases where moral standing would be attributed to non-conscious agents, if those cases took place in the real world. This suggests that the view that some forms of agency can ground moral standing has some intuitive appeal.

I cannot think of other kinds of cases which clearly support or weaken the argument for agency as a sufficient ground of moral standing. Therefore, we have reached a similar result in both considering theoretical principles as well as particular cases. Both kinds of considerations give weak to moderate support to the view that agency is a sufficient ground of moral standing, without being (at all)

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conclusive. To move the discussion of non-conscious grounds of moral standing forward, it is very helpful to also consider another source of arguments: theories of wellbeing.

5.3 Non-conscious suffering and wellbeing

Back in Chapter 2, we defined wellbeing as *non-instrumental goodness for*. Someone has a high wellbeing when their life is going well for them, non-instrumentally and all-things-considered. Theories of wellbeing attempt to specify the grounds of wellbeing, i.e., the things in virtue of which someone has the level of wellbeing they have. These are also called "welfare goods". In Chapter 2, we argued that being a welfare subject, i.e., being capable of wellbeing, having moral standing, and being capable of suffering are all equivalent. This suggests a simple method for investigating whether some beings have moral standing: assess whether they have some welfare goods, according to plausible theories of wellbeing. Following this train of thought, to know whether non-conscious beings have moral standing, we would only need to evaluate whether non-conscious beings can be in states which ground wellbeing, according to our best theories of wellbeing.

This is the method we will apply for the rest of this section. However, this method makes an assumption which needs to be defended first. This assumption has also been discussed by Goldstein and Kirk-Giannini (2025), who engage in the same type of project as I have suggested: that we 116can infer whether AI systems are welfare subjects and ultimately matter morally from theories of wellbeing. They call this assumption "Simple Connection". According to Simple Connection, an individual is a welfare subject just in case it is capable of possessing one or more welfare goods.

This assumption entails that theories of wellbeing are prior to accounts of welfare subjecthood: Theories of wellbeing specify which items are non-instrumentally good for one. If a being possesses some of these things, then it is a welfare subject. The competing view is that there are independent constraints on being a welfare subject. One such constraint might be being sentient. According to this view, a being only has wellbeing if it satisfies such constraints on being a welfare subject (e.g., being sentient) in addition to the possession of welfare goods, as specified by the correct theory of wellbeing. If this alternative is true, and Simple Connection is false, then the fact that a non-conscious being possesses welfare goods, as specified by the correct theory of wellbeing, does not imply that it is a welfare subject and has moral standing, since being a welfare subject might require consciousness or sentience in addition.

Following Goldstein and Kirk-Giannini (2025), I am sympathetic to Simple Connection. As the name suggests, Simple Connection is a more parsimonious view than the alternative. While Simple Connection posits only one condition for having wellbeing, namely, possessing welfare goods, other views posit two conditions regarded as independent, namely, satisfying conditions for the possession of welfare goods (as described by a theory of wellbeing) and being a welfare subject. More importantly, views which reject Simple Connection are unpleasantly ad hoc (van der Deijl 2021). They cannot satisfactorily explain why, if X constitutes wellbeing in sentient beings, the presence of X does not constitute wellbeing in non-sentient beings.

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To add another argument for Simple Connection, Goldstein and Kirk-Giannini (2025, sect. 6) construct a case where rejecting Simple Connection in favor of an additional consciousness requirement according to which a being has to be conscious in order to be a welfare subject gives implausible results. One can imagine a non-conscious paradigmatic agent which has many satisfied and frustrated desires. Moreover, suppose desire satisfaction is a welfare good. According to the consciousness requirement, the non-conscious paradigmatic agent would nevertheless not be a welfare subject and not have moral standing. However, if this agent were to consciously experience a single persistent visual experience of a homogeneous white field, then the consciousness requirement would be fulfilled, it would be a welfare subject, and the satisfaction of its desires would contribute to its wellbeing. It is implausible that such a conscious experience could change whether the satisfaction of desires contributes to wellbeing or not.[13] 117The case suggests, as Goldstein and Kirk-Giannini (2025) conclude, that whatever effect consciousness has on wellbeing "must be connected to individual welfare goods, rather than some extra condition required for wellbeing: rejecting Simple Connection is not well motivated".

There is a challenge to Simple Connection worth raising. Namely, on views on which companies and fictional characters have mental states (List and Pettit 2011), companies and fictional characters may be welfare subjects. This appears like an absurd implication. Goldstein and Kirk-Giannini posit an extra condition (the "reality constraint") to avoid this result.

However, I do not think any extra assumption is necessary to argue that fictional characters do not have beliefs and desires in the sense which is relevant. On most views, beliefs are mental states which, maybe among other things, play the right kind of causal role, being caused by perceptions and other beliefs and causing other beliefs and behavior. But fictional characters' putative beliefs do not *really* have a causal role; they only have one *within the fiction*. But, within the fiction, fictional characters often have moral standing too. The story of someone who mistreats a being with agency, beliefs, and desires is arguably a story where someone is immoral because they mistreat someone with moral standing. So, the view on which conscious is not necessary for moral standing arguably gives the correct verdict about fictional characters.

Groups, such as companies or nations, need a different treatment. We sometimes say of groups that they "believe" or "want" certain things, so there is some prima facie pressure to attribute mental states to them. It is possible to endorse an account of beliefs and desires which does not attribute them to groups but might attribute them to near-future AI. The right kind of functionalist account, on which the causal roles necessary for propositional attitudes such as belief and desire are more specific than what everyday use of terms such as "want" tracks, might suffice here. This is because, when giving a detailed psychological account, beliefs and desires in humans and other animals may turn out to function very differently than analogous states in companies.

In fact, the scientific study of belief examines questions such as whether belief-acquisition tends to be automatic or voluntary, whether beliefs are stored in one unified web or whether beliefs are stored inconsistently and redundantly, and which structure implicit beliefs may have (Porot and Mandelbaum 2021). It is not clear how one could even begin to collect relevant empirical data on the beliefs of

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companies, while such questions about information storage and acquisition can be answered with respect to AI systems. This suggests that the detailed causal role of human beliefs may be closer to the beliefs we may find in AI than in companies.

118So, I presuppose Simple Connection in what follows. This means that a being is a welfare subject if it instantiates some welfare goods. If so, we can look for theories of wellbeing to learn what properties beings must have to instantiate welfare goods. One may still worry that this top-down methodology (from theories of wellbeing to judgments about the welfare of individuals) is too simplistic, for example, because theories of wellbeing are controversial or may not generalize to all possible welfare subjects. However, it is eminently plausible that theories of wellbeing are at least one significant constraint on attributions of welfare subjecthood and there is no obvious constraint which rules out near-future AI wellbeing (see also the discussion at the end of Section 5.4). So, everyone should give the following considerations some weight.

The literature distinguishes three dominant families of theories of wellbeing. *Hedonists* claim that wellbeing is constituted exclusively by valenced conscious experiences, i.e., experiences which feel good or bad (Bramble 2016; Crisp 2006; Moore 2019). On hedonist views, the best lives are the ones that contain as many blissful experiences, and as few unpleasant experiences, as possible. Contra Goldstein and Kirk-Giannini (2025), I hold that hedonism straightforwardly rules out the possibility that non-sentient beings could have wellbeing. While there are accounts which understand pleasure in not consciousness-involving terms, hedonism still anchors wellbeing in pleasurable *conscious experiences*. Thus, according to hedonists, pleasure without valenced consciousness is not possible.

According to *desire-fulfillment theories*, wellbeing is determined by the extent to which desires (weighted by their strength) are fulfilled (Bruckner 2010; Heathwood 2016).[14] It is (non-instrumentally) good for you if your desires, particularly your strong desires, are satisfied. On this view, the best lives are those in which all of one's strong desires, such as overall life goals but also desires related to short-term material comfort and bodily integrity, are fulfilled to the highest extent. Notably, the constituents of wellbeing are not the positive feelings that may arise from the satisfaction of desires; otherwise, the theory would collapse into a version of hedonism. Instead, the constituent is the experience-independent fact that certain desires have been fulfilled.

The last candidates are *objective list theories* (Fletcher 2013; Griffin 1988; Lin 2014). According to such theories, whether your life is good for you is determined by whether it contains some subject-independent list of goods. For instance, a good life may require friendship, play, knowledge, excellence, and pleasure (Rice 2013). On such a view, the best lives are the ones which contain things which are objectively valuable. Objective list theorists hold that some catalog of features contributes to wellbeing 119irrespective of whether the subject desires them. In contrast to hedonistic views, these features are regarded as consisting not only in types of conscious experience.

It seems that non-conscious paradigmatic agents have wellbeing if the desire-satisfaction or the objective-list theory is true. For such agents can have desires, if we equate them with a substantive notion of goal, and apparently also goods like knowledge and perhaps friendship. There is a credible objection which can be made here. Theories of wellbeing have been developed almost exclusively with humans in mind. The aim to understand human wellbeing is what motivated theorists initially, and cases

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used to evaluate the adequacy of theories of wellbeing, e.g., the experience machine, are almost exclusively cases centering on humans. This might mean that these theories are not apt to capture non-human wellbeing.

In particular, these theories might be too permissive when considering AI systems. The worry is this: There might be certain properties which are necessary for wellbeing but not mentioned in any theory of wellbeing, because these properties are shared by all humans. For this reason, they might not have mattered to the debate on human wellbeing and are just a tacit background assumption of these theories. This would parallel the situation we find ourselves in with respect to theories of consciousness. According to some straightforward interpretations, due to the reason just mentioned, most prominent theories of consciousness implausibly imply that small networks comprising fewer than ten units (rough analogues to neurons) are conscious (Doerig et al. 2021; Herzog et al. 2007). Consciousness researchers typically respond by not taking these predictions of theories at face value, and by aiming to refine them (as we did in Chapter 4). Analogously, a strategy of directly applying theories of wellbeing to AI systems might lead astray.

For one specific way in which theories of wellbeing might turn out to be more demanding than a simple outline of them suggests, consider Lin's (2021) argument. He suggests that the satisfaction of the kinds of desires and objective goods which are welfare goods might require consciousness. He relies on Heathwood's (2019) view that the desire-fulfillment theory concerns only desires which are *genuine attractions*. Such desires involve viewing the object of the attraction with enthusiasm, pleasure, or excitement. Genuine attractions contrast with desires which are understood purely behaviorally, in terms of their role in motivating certain behaviors.

Based on this characterization, genuine attraction requires certain kinds of emotion-like states which may in turn require consciousness (see also Mogensen 2024). The same might apply to goods on the putative objective list (Lin 2021). Friendship is an example. Friendship may require desires in the genuine attraction sense, e.g., the desire to spend time with one's friend. If so, it might presuppose consciousness. Hence, Lin's argument entails 120that some wellbeing theories within all three big families of theories are compatible with the view that wellbeing requires consciousness. At the same time, he can also subscribe to Simple Connection.

Overall, I think that consideration of theories of wellbeing gives moderate support to the view that non-conscious beings can have moral standing. This is because desire-fulfillment and objective list theories are more popular in the philosophical literature than hedonist theories, and because a straightforward reading of these theories, in conjunction with Simple Connection, suggests that wellbeing does not require consciousness. However, as shown, one can reasonably object that the kinds of desires and objective goods which contribute to wellbeing require consciousness. This matter is not settled.

I conclude from this that we cannot be very confident in any view on whether non-conscious beings can have moral standing. In my view, our credence in this view should – in either direction – not be very far from 50%. Under moral uncertainty (MacAskill et al. 2020), this provides us with ample reasons to take the possibility of non-conscious suffering seriously. As a fictional example, suppose that an appropriate credence that non-conscious suffering is possible is 50% and that the probability that near-future AI systems in fact possess the properties which, conditional on the possibility of non-conscious

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suffering, suffice for the capacity for non-conscious AI suffering is 50% too. This would imply that, even if near-future AI systems are not sentient, the probability that they are capable of suffering is 25% (50% multiplied by 50%). Given that such suffering risks have a very large scale, a 25% probability would be extremely significant.

To make the case that suffering risks from non-conscious near-future AI systems are significant, we need to make the case that there is a significant chance that near-future AI systems will have properties which may ground non-conscious suffering. This is the task of the next section.

5.4 Assessing non-experiential AI welfare

In accordance with the methodology of the last chapter on AI sentience, I use current language models as a reference point for our investigation. When we think about near-future AI systems, we will picture them as advanced versions of current LLMs. LLMs are a general-purpose technology which can give rise to many different applications. Thus, if near-future versions of LLMs have moral standing, then a huge number of individual AI systems with moral standing will probably be created. Given the previous sections, to assess the probability that non-conscious near-future AI systems have moral standing, we need to answer three questions: How likely is it that such systems (i) are agents (in the sense specified in Section 5.2), 121(ii) have desires (in the sense of the desire-fulfillment theory of wellbeing), and (iii) have objective goods which might constitute wellbeing?

As pointed out by Goldstein and Kirk-Giannini (2025), so-called “language agents” are AI systems which may be particularly good candidates for attributions of moral standing. Since an excellent, more extensive characterization of language agents is given by Goldstein and Kirk-Giannini (2023), I will limit myself to the essentials. Language agents are created by combining an LLM with an agent architecture. The LLM performs the system’s cognitive processing, while the agent architecture arranges the individual processing tasks so that they result in coherent behavior over longer timescales. I will focus on the language agents described by Park et al. (2023). Their architecture has three main components (Park et al. 2023)[15]:

> The first is the memory stream, a long-term memory module that records, in natural language, a comprehensive list of the agent’s experiences. A memory retrieval model combines relevance, recency, and importance to surface the records needed to inform the agent’s moment-to-moment behavior. The second is reflection, which synthesizes memories into higher level inferences over time, enabling the agent to draw conclusions about itself and others to better guide its behavior. The third is planning, which translates those conclusions and the current environment into high-level action plans and then recursively into detailed behaviors for action and reaction. These reflections and plans are fed back into the memory stream to influence the agent’s future behavior.

So, the memory records natural language sentences which function as the language agent’s beliefs, desires, and observations. Then, these sentences are processed to draw relevant inferences about the language agent itself. Subsequently, the conclusions of these inferences are used to generate, first, high-

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level action plans and then detailed behaviors. The actual cognitive processing, which consists in generating natural language sentences from other natural language sentences, is done by the LLM. However, the "architecture of a language agent gives these sentences their functional roles by specifying how they are processed by the LLM in determining how the agent acts" (Goldstein and Kirk-Giannini 2023).

In the study by Park et al., language agents populate a small, isolated virtual environment (similar to the Sims) where they show believable imitations of individual human and emergent social behavior. In one example, once it is specified that one language agent intends to throw a Valentine's Day party, the agents independently distribute invitations to the party, make dates with each other, and coordinate to take part in the party at the 122same time. In this way, at least intuitively, language agents are much more like humans, who have beliefs and show typical behaviors, than pure LLMs.

However, to tackle the question about AI agency systematically, we need to ask whether some near-future AI systems, e.g., language agents, (a) possess goals which they robustly pursue across a range of situations, (b) behave in ways indicating a negative evaluation of the situation when their goals are not fulfilled and a positive evaluation of the situation when their goals are fulfilled, and (c) reflectively endorse their goals.[16] We said that many current AI systems are minimally goal-directed. This is especially obvious in the case of language agents. We gain predictive power over language agent behavior by treating the sentences in their memory stream as their beliefs and desires which enables us to, e.g., predict that they will show up at a Valentine's party at a specific time.

This mode of prediction and explanation works so well because language agents are designed such that their internal structure corresponds to humans' intuitive conception of psychology (folk psychology), which explains and predicts behavior by attributing beliefs and desires. In a minimal sense, a system like ChatGPT probably also has goals. We stated earlier that we are better able to predict its behavior when we interpret it as trying to give answers that humans find harmless, helpful, and correct.

When language agents have goals, like going to a specific party, they can take many actions over a relatively long time span to achieve these goals. Despite other observations and reasoning processes which occur in the meantime, they do not forget their goals. Moreover, their behavior can be made coherent so that they verbally express disappointment when their goal is frustrated or satisfaction if it is fulfilled. Again, the LLM's processing in conjunction with the overall architecture ensures that language agents respond in ways which are consistent with human folk psychology. Still, one could argue that genuine agents need to have long-term goals which they pursue for years and that they need to make complex multi-step plans in the service of these goals.[17] Hence, if the conditions are made demanding enough, language agents do not fulfill them. It is unclear how to tell how demanding the conditions for agency are or should be.

A further difficult question is whether language agents can reflectively endorse their goals. There is a case to be made for a positive answer. The reflection component of the language agent aims to make it capable of reasoning about its own beliefs and goals and to revise or maintain them according to the results of that reasoning. This is not only metacognitive but also seems to capture the precise functional role reflection is supposed to play in humans. One could deny that language agents are full-fledged agents by claiming that they lack beliefs, and that genuine 123reflection requires belief. However, again,

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the functional role of natural language sentences in the language agent's memory stream corresponds relatively closely to the role belief is supposed to play (Goldstein and Kirk-Giannini 2025). These sentences are based on observation and inference, used in further reasoning, and give rise to action plans. So, it appears that the conclusion that language agents are genuine agents is forced upon us.

I see two possible strategies for resisting this conclusion. It is hard to deny that the architecture of a language agent seems to have the right kind of structure for an agent. Thus, both strategies claim that the cognitive *processing* within the agent – performed by the LLM – is not sophisticated enough or of the wrong kind to count as agentic. First, one may doubt that the language agent's reasoning abilities, as expressed in their outputs, suffice for the capacity for genuine reflection. In effect, this means that the underlying LLM does not reason or does so only to an insufficient degree.

According to a strong version of this view, LLMs do not reason at all, but are merely "stochastic parrots" (Bender et al. 2021) which copy patterns they find in their training data. However, even if the processes within LLMs are "parrot-like" in some respect, LLMs can nevertheless sometimes produce texts of high quality about topics which are not contained within their training data (e.g., Piantadosi 2023). The same goes for drawing conclusions, solving puzzles (Bubeck et al. 2023; Lewkowycz et al. 2022; OpenAI 2023; Suzgun et al. 2022), etc. So, language models do not merely memorize responses from their training data (although this may explain a sizable chunk of their apparent linguistic competence; see Millière and Buckner 2024); they also come up – to some extent – with novel but still adequate responses. So, LLMs at least have abilities which imitate human reasoning.[18] That being said, there are also limitations. For instance, Arkoudas (2023) presents a number of apparently simple reasoning tasks that GPT-4 failed and concludes that "GPT-4 at present is utterly incapable of reasoning".[19] Overall, the overt reasoning abilities of current language models are impressive, but at least in some important respects inferior to human reasoning.

Alternatively, one could measure LLM reasoning not based on their overt abilities, that is, their behavioral outputs, but the underlying processes. Perhaps language models can produce outputs identical to what humans produce in many reasoning tasks, but they still only *imitate* human reasoning. The underlying process, so the worry, might not correspond to genuine reasoning. To explore this claim, it is necessary to learn more about the internal representations and algorithms the model employs, which is a nascent research area.[20] Currently, one may dispute whether research will support the claim that current LLMs can reason.

124Two lines of research seem particularly relevant: First, some studies aim to examine whether LLMs reliably internally represent the truth of statements, even in many cases in which they do not output true statements. Some authors describe these studies as measuring whether LLMs have beliefs (Levinstein and Herrmann 2024) or a concept of truth (Burns et al. 2022). While this seems premature, studies which aim to track internal representations which correspond to what the model represents as true could eventually provide us with a better sense of whether these models should be considered reasoners (Chalmers 2025).

Second, researchers have started to explore whether current LLMs have world models where a "world model" can be conceived of as an internal representation whose structure corresponds to the structure of the represented domain in the external world, and which is consulted in determining how to behave

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toward entities in this domain (Shagrir 2012). In the case of language models, an LLM would exploit a world model to decide which text to output in response to a given input. There is some tentative experimental indication that LLMs might have world models in this sense (Li et al. 2023; Meng et al. 2023; Yildirim and Paul 2023), but further research is needed.

Overall, there are some good reasons to say that even current language agents are genuine agents. They are obviously minimally goal-directed, they pursue goals robustly and consistently in spite of changing circumstances, and they plausibly reason about their goals. It is plausible that there is no difference in kind between the capacities of language agents and the capacities necessary for genuine agents. One may, however, maintain that there is a difference in degree. This may be either because goal-pursuit in language models is not robust enough or because their cognitive capacities do not suffice for genuine reasoning, either measured in terms of the quality of the models' outputs (particularly on inputs they have not been directly trained on) or measured in terms of their internal processing.

There are, however, two limitations to this strategy. First, LLM capability is bound to improve. It seems clear that future LLMs will be better at producing impressive textual outputs, including reasoning. In fact, a recent breakthrough has been made with the use of so-called "reasoning models", where models produce chains of reasoning relevant to solving a problem and are then trained via reinforcement learning on these reasoning chains (Xu et al. 2025).

It is less clear that the internal processes of LLMs will become more reasoning-like over time if they do not constitute genuine reasoning at the moment. However, it is still plausible that this will happen. After all, when LLMs are trained to become better able to solve reasoning tasks, their internal processing must somehow explain their behavioral performance. 125If so, the training might cause them to have more genuinely reasoning-like processes. Second, intellectually demanding accounts of agency, like the one needed to preclude language agents from counting as genuine agents, rule out that non-human animals and maybe human children, as well as some humans with severe mental disabilities, are agents. If one thinks that agency is morally significant, this might be an unpleasant implication (Jaworska 2007b).

Do language agents satisfy the conditions for having wellbeing? First, we need to ask whether they have the kinds of desires that the desire-fulfillment theory specifies as relevant. The relevant factors parallel the previous discussion of agency. The natural language sentences stored inside the language agent's memory stream mirror the functional role human folk psychology ascribes to desires. They explicitly contain information about the agent's goals, they are refined through further processing, and they are used (in conjunction with other sentences which resemble beliefs) to form plans for action. Since they are explicit language sentences whose content can be true or false (e.g., "go to a party tomorrow evening"), it is easy to recognize whether they have been fulfilled or frustrated.

At the same time, one could postulate a very demanding notion of desire. Beliefs and desires are often treated analogously in philosophy, and they interact closely. So, if having a world model or an internal representation of truth turns out to be necessary for belief, then it is plausible that it will be necessary for the possession of desires as well.[21] It is uncertain whether near-future LLMs, and thereby the language agents dependent on them, satisfy these conditions.

Moreover, analogously to our treatment of agency, we exclude from consideration here notions of desire which presuppose that only conscious beings can have desires. However, for a full picture, we

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need to note that such conceptions exist. For instance, Schwitzgebel (2002) champions a *wide* dispositionalist account of belief which holds that beliefs are individuated not only by their causal relations to behavior and non-conscious mental states but also by their dispositions to be caused by and to cause certain conscious experiences. Recall also Lin's (2021) view on wellbeing which belongs in the same category. According to him, it is plausible that only certain kinds of desires matter for wellbeing and that these kinds of desires presuppose consciousness. Finally, Smithies (2019) argues that justified belief and knowledge require consciousness which may also restrict the range of welfare-relevant desires that non-conscious creatures can have.

Hence, there are some moves one can make to reject the view that current language agents are welfare subjects, according to the desire-fulfillment theory of wellbeing. Nevertheless, if one does not explicitly tie desire possession to consciousness, these strategies gradually become less plausible 126when the capability of LLMs increases and language agents approach paradigmatic non-conscious agents. Thus, if wellbeing does not require consciousness and the desire-fulfillment theory is true, then it seems likely that language agents will be welfare subjects in the future, maybe even in the near future.

Can language agents have welfare goods which are described in the objective-list theory? Similar reasons apply. Knowledge is one potential welfare good. If language agents have beliefs, then they are also capable of knowledge, since they can reliably form true beliefs and provide justifications for their beliefs (Goldstein and Kirk-Giannini 2025). While it seems likely to me that at least near-future language agents have beliefs, the previous caveats apply: belief, or at least the form of knowledge required for wellbeing, might presuppose conscious experience and there is a chance that belief possession has demanding cognitive requirements that language agents won't even fulfill in the near future.

Another prominent welfare good is friendship. Can language agents have friends? According to most theories of friendship, a necessary condition for friendship is that friends care about each other or feel a particular kind of love toward each other (see the references in Helm 2023). Helm (2023) claims:

> [T]here is widespread agreement that caring about someone for his sake involves both sympathy and action on the friend's behalf. That is, friends must be moved by what happens to their friends to feel the appropriate emotions: joy in their friends' successes, frustration and disappointment in their friends' failures (as opposed to disappointment in the friends themselves), etc.

This suggests that friendship requires certain emotions. If the relevant kinds of emotions essentially involve conscious experiences, then friendship likely requires conscious experiences.[22] If so, language agents can only have friends if they are conscious. Moreover, it seems plausible that friendship demands a long-term commitment to another person. One cannot be friends with someone, then suddenly stop being friends, and then again be friends one day later, and so forth. Arguably, language agents – as of now – lack the consistency to behave like a friend robustly and reliably for months and years, in spite of other events influencing them. So, it is doubtful that current language agents can have friends. Even if

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near-future language agents can behave more like a friend, friendship still likely presupposes consciousness. If so, then friendship is not relevant to the moral standing of language agents. If they are conscious, then they likely have moral standing anyway. If not, then they cannot have friends.

127Excellence is another candidate objective welfare good. It is contested what exactly the relevant form of excellence consists in (Wall 2021). A widely shared claim is that it is connected to maximum "achievement of human excellence in art, science and culture" (Rawls 1971, p. 325). There are two ways to make sense of this claim in the context of AI. First, we might say that excellence consists, for all beings, in achieving things in the domains of science, art, and culture which are at or above the maximum which is possible for humans. For instance, AI has surpassed human capabilities in chess and protein folding (Jumper et al. 2021). Thus, according to this notion, it possesses this welfare good. Alternatively, we could say that excellence is relative to the kind of being in question. For instance, Dorsey (2010, p. 61) says that:

> in principle perfectionism could be applied to any creature. The best life for a cat depends on the sort of creature a cat is – developing and exercising those capacities that make a cat a cat is what makes for a good cat life.

If this is true, it seems that – by definition – any being can realize excellence. Language agents are excellent if they develop and exercise the capacities that make them what they are. By definition, they must be capable of doing that. So, this way of understanding excellence would also imply that AI systems have wellbeing. However, it seems to entail that *every* creature has wellbeing, which undermines the significance of wellbeing attributions to AI in the first place. Plausibly, this is a sign that this way of understanding excellence should be rejected. If one wants to reject the consequence that all creatures are welfare subjects but still hold on to the view that which capacities constitute excellence is relative to the kind of creature, one needs to appeal to an independent standard to determine which beings can have the capacities whose exercise constitutes a good life. In this case, thinking about excellence does not help us to determine whether language agents have wellbeing.

We will now discuss an important objection which applies to the whole project of this chapter, and arguably also of previous chapters. Some researchers might object that attributions of agency or wellbeing to language agents are simply absurd. If certain theories of wellbeing or agency imply that language agents have moral standing, they might interpret this as a reason to reject these theories. They may conclude that, e.g., the desire-fulfillment theory is flawed or that it needs to be supplemented with an additional constraint for the possession of morally relevant desires which language agents don't satisfy. I cannot rule out that this response is correct. Nevertheless, there are good reasons to resist this reductio ad absurdum argument.

128First, let us note that this response is less attractive in the context of attributions of AI sentience. Almost everyone agrees that sentience is sufficient for moral standing. Thus, in the case of sentience, this reductio would take certain theories of consciousness or valence to be false because they suggest near-future AI sentience. Thus, this objection would amount to disbelieving *scientific* theories on the basis that one wants to maintain a preferred *ethical* conclusion. This seems like an epistemically illegitimate form of wishful thinking. One should not reject scientific results and hypotheses *merely*

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because they have ethical implications which one is skeptical of or dislikes (Impagnatiello forthcoming; Timmerman et al. 2024).

By contrast, in the context of AI wellbeing and agency accounts of moral standing, one would reject an ethical theory because it – in conjunction with empirical observations – leads to an ethical conclusion one rejects. This is less clearly illegitimate. Arguably, it is a legitimate application of the method of reflective equilibrium (Daniels 2020). This method advises us to make our general ethical principles (e.g., agency determines moral standing) and our ethical judgments about specific cases (e.g., robot X has moral standing) coherent, by adjusting either the general principles or the judgments about specific cases.

However, one needs some reason to justify the claim that attributions of moral standing to AI are absurd. It is not permissible to just reject conclusions one does not like without any arguments in support. It is hard to see what this reason could be. Near-future language agents are not obviously beings without moral standing. In other words, it is not absurd to attribute moral standing to them. We have argued for the view that there may be moral standing without sentience. So, the absence of sentience (which is also uncertain) does not imply that near-future language agents lack moral standing.

The most influential theories of mental states, especially of mental states other than consciousness, are functionalist or dispositionalist. That is, they imply that a being can have mental states purely in virtue of having the right kind of internal causal organization and processing or displaying the right kind of behavior, independently of what material they are made of or whether they exist only virtually. In virtue of their underlying LLMs, language agents perform quite complex cognitive processing and behavior. For this reason, the mere fact that language agents are not biological but move in a virtual environment and are ultimately realized in silicon-based systems is not a sufficient reason to deny that they have mental states which could ground moral standing either.

Abstracting from these putative differences in sentience and physical realization, near-future language agents appear as a very human-like life-form. They are intelligent, communicate in language, and behave in accord 129with humans' intuitive conception of psychology. If such beings were to inhabit human-like physical bodies, including human facial expressions, it seems plausible that it would appear pretty intuitive that they have moral standing and that it is wrong to harm them without a reason. This suggests that, at root, bias against beings which look very different from us may be what prevents us from taking the possibility of near-future language agents with moral standing seriously. Since humans may have self-serving and speciesist biases against AI welfare, it is especially questionable to reject the conclusion that near-future AI systems might suffer without a proper argument.

Another way to express the previous points is that several sources of evidence converge on the conclusion that near-future AI moral standing is a serious possibility: theories of sentience, theories of wellbeing, agency accounts of moral standing, accounts of mental states, and a consideration of the intuitive judgments that would in fact be elicited if we encounter the right kinds of near-future AI systems. In such a case of convergent evidence, the judgment that near-future AI suffering is absurd lacks much support: it is a mere dogmatic reaction. Given the reasons for bias especially, it would be unwise to attribute much significance to it.

Practically speaking, it is especially morally hazardous to attach overriding weight to absurdity judgments when making high-stakes decisions. This is because we cannot properly estimate the risk of error of such judgments. When basing judgments about AI suffering on philosophical and scientific

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evidence and explicit reasoning, then we inevitably make the assumptions about AI suffering risk explicit, on which our views rest. This has practical benefits: We can use these theories and arguments, and explicitly considered reasons in favor of and against these theories, to inform appropriate credences in the relevant assumptions and in AI suffering risk in general. This way, we gain a sense of which uncertain assumptions our assessments rest on, how they might be flawed, and what the overall risk of error is. Moreover, we can notice when further evidence for or against relevant assumptions should make us adjust our judgments about AI suffering risk.

By contrast, if our judgments about AI suffering risk are mostly determined by blanket absurdity judgments, both these advantages are missing. First, mere absurdity judgments are not based on any explicit arguments or assumptions, so it is not possible to estimate how confident we should be in them or what the risk of error is. For the same reason, it is unclear – in this scenario – how to update on new information: It is open which degree of evidence should make one revise one's viewpoint. So, since absurdity judgments are not epistemically transparent – it is opaque what their degree of justification is – they encourage behavior which is not sufficiently sensitive 130to uncertainty about the subject matter and to newly incoming information. So, they invite morally reckless decision-making under uncertainty.

Finally, the fact that all reasonable interpretations of leading theories of wellbeing, sentience, and moral standing suggest in combination that there is a significant chance that near-future language agents can suffer surely has a *non-negligible* evidential weight, even if opportunities for doubt remain. In a situation where many reasons, perhaps the overall weight of reasons, support believing in near-future AI suffering, it would be morally reckless to be very highly confident that near-future AI is not capable of suffering, without having substantive reasons in favor of this view. A reductio argument, which is not backed by good independent reasons in support, cannot justify a credence lower than 1% in near-future AI suffering.

5.5 Summarizing the prospects of near-future AI suffering

It is time to take stock and reflect on what the previous discussion suggests (see Table 5.1). In Chapter 4, we have concluded that, given current evidence, there is at least a 1% chance of near-future AI sentience. The arguments of this chapter further increase the chance that near-future AI is capable of suffering. Due to the large space for reasonable disagreement, I won't defend a specific credence for the view that near-future AI will suffer in virtue of states other than conscious experiences. I will only say that the previous argument suggests that the probability is not trivial.

TABLE 5.1 Indicators of non-conscious suffering Return to text.#

Indicators	*Evidence for near-term AI suffering*
Agency as involving robust goal-pursuit and reflection	Moderate evidence
Desire-fulfillment theory of wellbeing	Moderate evidence
Objective-list theory of wellbeing	Moderate evidence

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Note: This table depicts the three indicators of non-conscious suffering that the text argued for and how much additional evidence of near-term AI suffering they provide. It is concluded that they all provide moderate evidence, namely, evidence which is significant but also quite uncertain.

We have seen that there are credible arguments, having significant intuitive pull, according to which non-conscious agents have moral standing. In addition, a straightforward, albeit somewhat controversial, reading of two of the three leading theories of wellbeing entails that non-conscious beings can have moral standing. Due to this, there are strong reasons why near-future language agents might have moral standing, even if they are not sentient. That being said, one can still reasonably reject this conclusion. 131 This is because one may subscribe to demanding accounts of what it takes to have beliefs, desires, to pursue goals, and to reflect on them. I cannot rule out these views. They might well be true. At the same time, it is unclear why we should prefer them to more permissive alternatives. Not only that but also, as AI research advances, increasingly demanding conditions for moral standing will be satisfied by language agents and their underlying language models.

This chapter has shown that the threat of non-conscious suffering provides an independent reason to be concerned about AI suffering risks. For this reason, it should increase our estimate of the probability of near-future AI suffering, although I will leave open by how much. Under uncertainty, this means that AI suffering risk constitutes an urgent and important issue.

There are three further considerations which are important to note for appropriately contextualizing this assessment. First, the probability of creating near-future AI capable of suffering is not equal to the probability that astronomical AI suffering will be created in the near future. This is because it is a logical possibility that AI capable of suffering will be created but that no or few AI systems will actually suffer, or that only a few AI systems capable of suffering will be created. This does not appear likely if humans do not explicitly attempt to reduce AI suffering risk. Assuming that humans have no concern for AI welfare, it seems likely that AI systems will be treated in ways that are bad for them, if there is something which is bad for them. This is because, in this scenario, humans will train and deploy these systems in ways useful for humanity. There is no strong reason to believe that this lines up neatly with what is, and feels, good for these systems in most cases.

Moreover, given that very capable systems like LLMs and language models are capable of suffering, it is likely that they will be used by billions of humans in all kinds of ways. Language agents, for instance, can also process information in virtual environments without continual human prompting. So, only the available computing power limits the number of language agents which can exist. Given that the cost of computing power decreases over time, and absent concern about AI suffering, it seems likely that countless AI systems[23] will be used. Even a single human being, with enough computing power at their disposal, could create billions of AI systems capable of suffering. So, there is reason to think that, conditional on the existence of near-future AI capable of suffering, astronomical AI suffering is likely.

Importantly, though, all the previous estimates were conditional on the absence of dedicated human effort to not create AI suffering. The hope animating this book is that AI suffering risk will be recognized as a concern and that individuals and governments will take measures to reduce it. If so, 132

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this would reduce the likelihood that near-future AI systems capable of suffering will be created, that they will actually suffer, and that this suffering will be astronomical. Hence, I do *not* claim here that responses to AI suffering risk are unlikely to be effective.

Second, I have argued in the last chapter that even a 1% chance of AI suffering is morally significant. Letting someone else incur a 1% chance of high suffering without a good reason is immoral. It does not matter morally whether this chance concerns a being which is certain to be sentient or whose sentience we are uncertain of. The previous argument was based on empirical uncertainty, namely, uncertainty about the empirical question whether non-human beings have sentience. In this chapter, we are confronted with empirical as well as moral uncertainty. The questions which capabilities near-future AI systems will have, whether they will be agents (given a specific conception of the term), and whether they will have beliefs and desires, are empirically uncertain. By contrast, the question of whether agency, desires, or some items on the objective list suffice for being a welfare subject, and which form of these properties does so, is morally uncertain. It is uncertain because it is not clear which ethical view is correct. While there are ongoing disagreements (Gracely 1996; Gustafsson and Torpman 2014; Weatherson 2014), MacAskill et al. (2020) have presented compelling arguments according to which moral and empirical uncertainty should be treated analogously. If both kinds of uncertainty obey the same, or similar, principles, then uncertainty about the possibility of non-conscious suffering also gives us strong moral reasons to reduce AI suffering risk.

Third, not all kinds of suffering are equally strong. In humans, it seems to me that the strongest possible negative welfare states involve negative conscious experience. Consider immense physical pain (e.g., cluster headaches), torture, incapacitating fear, and mood disorders (e.g., major depressive disorder) – all of them would not be nearly as bad as they are if they did not involve negative conscious experience. As I have argued, it might be the case that even unconscious frustration of desires, for instance, constitutes suffering. This suffering might also be quite significant if it involves some desires which are very important to someone. For instance, if one has a strong desire that one's children are having a happy life, then the frustration of this desire can plausibly be very bad for one, even if one never learns that one's children are unhappy, such that the frustration of this desire never causes negative conscious experience.

Nevertheless, it seems to me that the extent of the negative welfare states constituted by the mere frustration of one's desires, if it is not connected to consciousness, cannot compare to the negative welfare states which are constituted by the most terrible conscious experiences humans can have. However, I will not argue for this claim here, since my larger argument does 133not depend on it. I will merely make a further point. Suppose this is not the case. That is, suppose non-conscious suffering can be as strong as the strongest conscious suffering. Nevertheless, beings which can have negative conscious experiences *as well as* non-conscious suffering are in danger of stronger forms of suffering than non-sentient beings since the latter can only be subject to one form of suffering. All of these ideas transfer to AI. For this reason, I think that AI suffering risks are particularly grave with respect to systems which might be sentient. While all AI suffering is important, we have especially strong moral reasons to reduce the risk that sentient AI systems suffer.

This chapter concludes my argument for taking AI suffering seriously. Such risks have – already in the near future – a significant probability of occurring. In conjunction with the astronomical scale of this

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risk and its relative neglectedness so far, this makes reducing AI suffering risk an urgent topic of consideration.

Notes

1. However, this does not need to remain the case. For example, suppose an AI system produces verbal reports which have been well-validated to reflect accurate introspection of its internal states (Perez and Long 2023). Such a system may, conceivably, report that it has motivating consciousness but lacks valenced experiences and desires, producing evidence for the former but not the latter. Return to text.#
2. For other relevant discussion, see Dung (n.d.), Moosavi (2023), and Shepherd (2023, 2024). Return to text.#
3. That being said, questions about which concepts we should use, and which conventions we should impose, can be philosophically significant, as advocates of conceptual engineering argued (Cappelen 2018). Return to text.#
4. Some people might be happy to endorse the view that bacteria have moral standing (e.g., Agar 2001; Taylor 1986). I disagree but will not make this case here. I will say, however, that a view according to which all living beings have moral standing will, in practice, diminish the importance of attributing moral standing to a being. In this scenario, we will be pressed to make practically significant distinctions between the degree of moral weight of different beings. On this view, one may view the project of this book as investigating whether AI systems can have a higher degree of moral weight, which is more akin to the moral weight animals or humans have, than plants or bacteria. Return to text.#
5. In computational terms, this distinction corresponds to the distinction between particular forms of model-based and model-free reinforcement learning, respectively. Return to text.#
6. However, there is ample evidence of goal-directed control in rodents (Balleine and O'Doherty 2010; Huh et al. 2009). Return to text.#
7. I ignore here the problem of deviant causal chains (Mayr 2011, ch. 5). So, propositional attitudes are – according to this proposal – not sufficient for agency, even if they cause behavior in non-deviant ways. Return to text.#
8. On this, see also Jonathan Birch's talk on the evolution of reasons: https://drive.google.com/file/d/1Yi9DdHGHjdZyxVrt1Z_altuOzlNv0u1e/view (Accessed May 1, 2023). Return to text.#
9. 134Of course, one can assume a very deflated sense of what it takes for interests to matter to a subject, so that the agency interest link obtains, even based on a minimal notion of goal. However, given this understanding, it becomes questionable that fulfilling the meta-criterion proposed by Kagan and Ladak is sufficient for moral standing. Return to text.#
10. If someone finds the talk of "mattering to a being" not compelling, one can elicit the same intuition by simply talking about "mattering". We have seen that not all minimal goals matter

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morally. There is no reason why being an instance of goal-directed control or having a long time horizon should constitute the crucial distinction between goals which matter morally and goals which do not. Return to text.#

11. One could further support the view that all and only these kinds of goals matter for moral standing by arguing that such goals are desires in the *genuine attraction* sense which is the sense of desires which matters for wellbeing, in contrast to purely behavioral desires, according to Heathwood (2019) (see also next section). Relatedly, one could argue that these are the goals that resonate with beings so that they are not "alienating" (Fanciullo 2025). Return to text.#
12. This observation is similar to Schwitzgebel's (2001) argument for in-between believing. Note that the assumption that goal possession requires certain behavioral dispositions may only be plausible in cases where the being has the behavioral *capacities* conducive to obtaining the goal but does not exercise these capacities. Return to text.#
13. The thought-experiment delivers the same verdict when we posit sentience as requirement on welfare subjecthood and imagine that the experience of a white visual field is slightly pleasant to the subject. It is still implausible that this experience makes a difference to whether desire satisfaction contributes to wellbeing. Return to text.#
14. On virtually all desire-fulfillment theories, there are some constraints on the kinds of desires which count for wellbeing (Heathwood 2019; Lukas 2010). For example, theorists want to rule out that the satisfaction of desires caused by compulsion increases a subject's wellbeing. Return to text.#
15. See also Wang et al. for an overview of the recent literature (Wang et al. 2024). Return to text.#
16. Recall also that we said that (b) might not be necessary for being an agent. Return to text.#
17. One may hold that objective time or represented (subjective) time are relevant here, but not experienced (subjective) time if language agents lack conscious experience. Return to text.#
18. Goldstein and Levinstein (n.d.) also review empirical evidence which undermines the view that LLMs only perform memorization. Return to text.#
19. Note that limitations found in older language models often do not generalize to state-of-the-art systems. Return to text.#
20. The background is that artificial neural networks are opaque (Beisbart and Räz 2022; Heuillet et al. 2021; Zerilli 2022). While the network architectures and the weights of the connections between units are known, the dynamics of the network is too complex to understand or explain the network on this level. Return to text.#
21. For further thoughts on how the functional role of human desires might differ from the states of current language agents, see Butlin (forthcoming, sect. 5). Return to text.#
22. Since there is credible evidence of unconscious perception (Michel 2023; Prinz 2015), I believe that unconscious emotion – functioning similarly – is also a serious possibility. At the same time, completely unconscious emotion may not be ethically significant, such that it does not suffice for friendship or desires in the genuine attraction sense (if they presuppose emotion in the first place). Return to text.#
23. A specific issue I will not go into here is how to count AI systems, and how this issue influences welfare assessments (Roelofs and Sebo 2024). If ChatGPT is used by a billion of users and any

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interaction causes it to suffer, then its suffering would likely be higher than that of any single human. Still, it is not correct to say that a billion ChatGPT's exist. Perhaps we could say that the general program ChatGPT is realized a billion different times. Return to text.#

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